



A Survey of Link Prediction in Temporal Networks

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Abstract

Temporal networks have gained significant prominence in the past decade for modelling dynamic interactions within complex systems. A key challenge in this domain is Temporal Link Prediction (TLP), which aims to forecast future connections by analysing historical network structures across a range of applications. While existing surveys have addressed specific aspects of TLP, they typically lack a comprehensive framework that distinguishes between representation and inference methods, the two key ingredients of TLP. This survey bridges this gap by introducing a novel taxonomy that explicitly examines representation and inference in existing methods, providing a novel classification of approaches for TLP. We analyse how different representation techniques capture temporal and structural dynamics, examining their compatibility with various inference methods for link prediction tasks. Our taxonomy not only clarifies the methodological landscape but also reveals promising unexplored combinations of existing techniques. This taxonomy provides a systematic foundation for tackling emerging challenges in TLP, including model explainability and scalable architectures for complex temporal networks.

Keywords Temporal networks · Dynamic networks · Link prediction · Graph learning · Machine learning

Introduction

A temporal network is a network whose structure and properties evolve over time. In the past decade, temporal networks have been used to model interactions between the components of complex systems with dynamic structures and characteristics. A temporal network consists of two key elements: a set of nodes representing components of a system and a set of links indicating interactions between pairs of nodes. In temporal networks, links between nodes may appear or disappear over time, and new nodes may also join. Broadly, temporal networks are represented in two

paradigms: snapshot-based and continuous-time (event-based). Snapshot-based networks discretise time into intervals, yielding a sequence of static graphs, which is intuitive but may lose fine-grained temporal information. In contrast, continuous-time networks preserve the exact timestamps of interactions (events), enabling models to capture dynamics at arbitrary resolutions [1, 2]. This distinction forms the foundation for most temporal representation learning approaches and is particularly relevant to temporal link prediction (TLP).

Identifying the likelihood of connecting between nodes to predict future links in a network (link prediction) is a fundamental problem that can be applied to various domains such as social networks [3], telecommunication networks [4], traffic forecasting [5, 6] or knowledge graphs [7, 8]. Link prediction was first discussed in the context of static networks [9] where the aim has been to predict future links based on a network's current state. In this paper this is referred to as traditional link prediction. However, TLP uses a network's historical states to predict the likelihood of future links. Incorporating historical dynamics into the TLP problem introduces additional complexity [10, 11]. In social networks for instance, such as the one shown in Fig. 1, the TLP task is to predict the likelihood of a future link between

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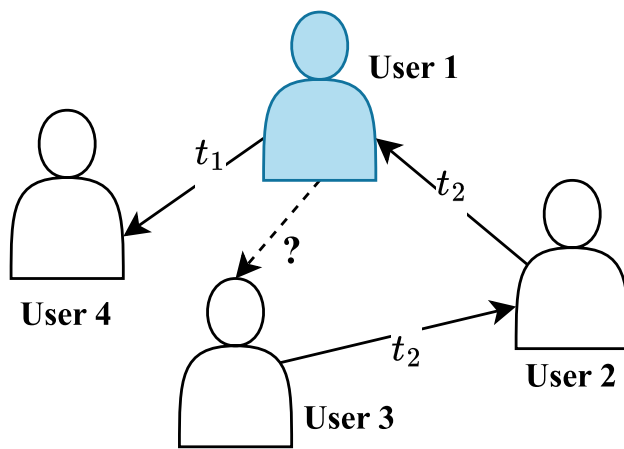


Fig. 1 Temporal link prediction in a social network

User 1 and User 3 based on the observed temporal interactions among users.

Recently, numerous methods have been proposed to address the TLP problem, each method focusing on different network characteristics. Solving TLP requires two key ingredients: first, a *representation unit*, which models how a temporal network can be captured to reflect its traits and historical dynamics; and second, an *inference unit*, which determines the likelihood of future links based on the learned representation. Since all TLP models necessarily need both units, this review is confined to approaches concerning either representation or inference.

Traditional surveys often categorise TLP models into broad methodological families such as “learning-based” [11]. This perspective provides a useful overview of existing approaches, but tends to emphasise overall model families rather than disentangling specific design choices. For instance, both TGN [1] and EvolveGCN [12] are typically labelled as “learning-based,” yet their underlying assumptions differ markedly: TGN employs a continuous-time message passing representation units with attention-based units making it effective for modelling of rapidly evolving interactions, whereas EvolveGCN adopts a sequence of discrete snapshots with RNN-based updates of GCN parameters, which is more suitable for gradually evolving networks.

In the paper, TLP models are classified according to whether their primary contribution is to representation or inference; the defining attributes in each unit form the basis for the taxonomy adopted here. By separating representation and inference units, our taxonomy highlights such distinctions more explicitly, thereby clarifying how models align with different data characteristics and task requirements, and revealing unexplored combinations. The survey begins by examining various representation methods and discusses their merits and challenges. It then classifies and evaluates the inference approaches that have been explored

in the literature. Guided by this methodological perspective, the discussion is subsequently divided into two sections “[Review of Representation Units for TLP](#)” and “[A Review of Inference Units for TLP](#)” dedicated to each unit, respectively.

The survey highlights potential research opportunities through two primary avenues: (i) unexplored combinations of existing representation and inference units and (ii) the development of novel unit designs that could introduce new computational paradigms for either representation or inference. These opportunities could lead to more effective TLP approaches.

Various surveys have been conducted to review TLP studies. Some surveys concentrate on specific network structures, such as homogeneous networks [13, 14], or directed networks [15]. Other surveys focus on the application of TLP across various domains, encompassing temporal knowledge graphs [16] and social networks [3, 17, 18]. Our survey sets itself apart in several aspects:

- The survey introduces a novel taxonomy that distinguishes between the representation and inference units. Unlike existing surveys [13–15] which classify models under a single hierarchical framework, our approach separates these components to provide deeper insights into the models’ applicability and performance variations whilst also exploring diverse network characteristics pertinent to the TLP problem.
- The survey provides a comprehensive review of the representation unit. While previous surveys [3, 17] provided some discussions of these models, we offer a systematic classification and analyse their ability to capture temporal and structural dynamics, thereby addressing a significant gap in the literature.
- The survey identifies current challenges and outlines potential directions for future research in representations, network characteristics, and TLP methodologies, offering valuable guidance for subsequent investigations in this field.

Methods included in this survey come from papers identified using Web of Science, Scopus, and Google Scholar and keywords on temporal networks, network representation and link prediction. Only articles written in English in the period 2000–2025 that propose relevant algorithms or methods were considered.

The rest of this paper is organised as shown in Fig. 2. “[Background](#)” presents the background on temporal graph representation and the problem statements. Following this, “[Taxonomy](#)” introduces the taxonomy framework. “[Review of Representation Units for TLP](#)” and “[A Review of Inference Units for TLP](#)” provide an overview of TLP methods,

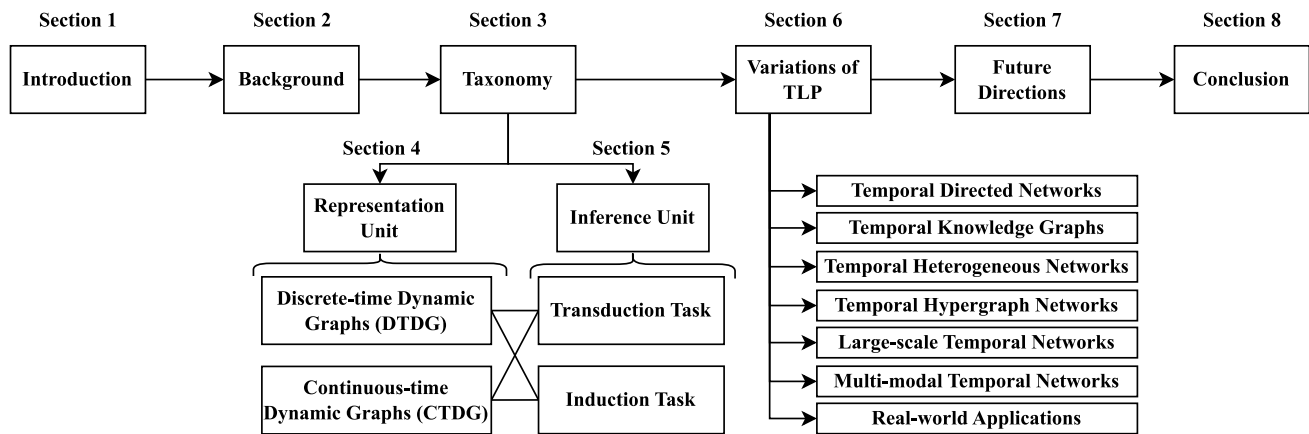


Fig. 2 Overview of the survey

classifying them based on their representation and inference units. “Variations of the TLP Problem” and “Future Research Directions” explore variations of the TLP problem and outline potential directions for future research. Finally, “Conclusion” presents the survey’s conclusion.

Background

Temporal Networks

A temporal network represents interactions between components in a dynamic system over time. A dynamic system comprises elements that interact with each other, and these interactions may change over time.

A temporal network can be modelled using a temporal graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{T}, \mathcal{X})$ [10], where $\mathcal{V}$ is a set of nodes, and $\mathcal{E}$ is a set of edges denoting the interaction between pairs of nodes. The notation $\mathcal{T}$ and $\mathcal{X}$ indicate a time domain and a set of nodes’ attributes, respectively. A temporal network can be defined as $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{T})$ if node attributes are ignored.

Problem Statement

Temporal Link Prediction (TLP): Given a temporal network $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{T}, \mathcal{X})$ and a current timestamp $\tau \in \mathcal{T}$, TLP aims to predict future edges formed between nodes in the set $\mathcal{V}$ after timestamp τ , based on the historical graph preceding τ .

Unlike traditional link prediction, which focuses primarily on predicting links using node attributes and current topology, TLP requires models that capture complex temporal and spatial network dynamics. This presents the challenging task of forecasting future link formation based on both historical and current network states.

Representation Methods

Temporal graphs can be described using two distinct approaches for the time dimension:

1. **Discrete-time Dynamic Graphs (DTDG):** A DTDG simplifies a temporal network into a sequence of time-stamped snapshots, converting the continuous temporal space into discrete timestamps. This allows dynamic graphs to be represented in a static, analysable format, though some information may be lost between snapshots during discretisation.
2. **Continuous-time Dynamic Graphs (CTDG):** This method abstains from discretising the time dimension. Instead, it characterises the presence of all elements independently within a continuous temporal space, preserving the inherent temporal continuity of temporal networks.

Discrete-Time Dynamic Graphs

Discrete-time dynamic graphs define a temporal network $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{T}, \mathcal{X})$ as a sequence of snapshots $\mathcal{G} = (G_1, G_2, \dots, G_{|\mathcal{T}|})$ over a discrete set of timestamps $\mathcal{T} = \{1, 2, \dots, |\mathcal{T}|\}$ where $|\cdot|$ is the size of the set. Accordingly, the sequence of edges $\mathcal{E}$, nodes $\mathcal{V}$ and corresponding attributes $\mathcal{X}$ can be expressed as $\mathcal{E} = (E_1, E_2, \dots, E_{|\mathcal{T}|})$, $\mathcal{V} = (V_1, V_2, \dots, V_{|\mathcal{T}|})$ and $\mathcal{X} = \{X_1, X_2, \dots, X_{|\mathcal{T}|}\}$. The interval between consecutive snapshots is assumed to be regular.

Each snapshot at timestamp t is represented as a tuple $G_t = (V_t, E_t, X_t)$, where $V_t = \{v_1^t, v_2^t, \dots, v_{N_t}^t\}$ denotes the set of nodes; N_t is the number of nodes at timestamp t . $E_t = \{((v_i^t, v_j^t), w_{ij}) \mid v_i^t, v_j^t \in V_t, w_{ij} \in \mathbb{R}_+\}$ denotes the set of edges and $((v_i^t, v_j^t), w_{ij})$ represents the edge (also called link or event) between node v_i^t and node v_j^t and may be associated with a weight w_{ij} . An unweighted graph is a

special case of a weighted graph with all edges weighting 1. Similarly, the attribute set X_t is $\{\phi(v_1^t), \phi(v_2^t), \dots, \phi(v_{N_t}^t)\}$ and $\phi(\cdot)$ is the mapping function. Note that the attribute formation is usually a vector $x_i^t \in \mathbb{R}^d$, and thus X_t becomes an attribute matrix $X_t \in \mathbb{R}^{N_t \times d}$.

Moreover, the topology of snapshots can be expressed by an adjacency matrix set $\mathcal{A} = \{A_1, A_2, \dots, A_t, \dots\}$. Here, $A_t \in \mathbb{R}^{N_t \times N_t}$ corresponds to the snapshot G_t . For undirected unweighted graphs, $A_t = \{a_{ij}^t\}_{i=1, j=1}^{N_t \times N_t}$ is a symmetric binary matrix. Conversely, for undirected weighted graphs, A_t equals the weight matrix whose elements are the weight of the edges.

Figure 3 provides an example of the discrete-time dynamic graphs for a temporal graph modelling an online social network. Each timestamp t features a snapshot G_t that depicts the system's behaviour, akin to user interactions in online social networks at a specific time. To minimise information loss, the interval between two adjacent snapshots should ideally equate to the minimum event duration. However, this can lead to significant data sparsity and redundancy in certain scenarios. For instance, if an email network is displayed as a series of snapshots, many snapshots might be empty or show very few connections [19]. Consequently, discrete-time dynamic graphs often entail a trade-off between information loss and representational efficiency, which needs to be tailored to the specific research demands. The fundamental premise is to depict temporal networks as a sequence of static networks.

Continuous-time dynamic graphs

Unlike discrete-time dynamic graphs, continuous-time dynamic graphs do not require equal time intervals between timestamps. For a period $T \subseteq \mathcal{T}$, a continuous-time dynamic graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{T}, \mathcal{X})$ can be expressed as $G_T = (V_T, E_T, T, X_T)$, where $\mathcal{T} = \{t_1, t_2, \dots, t_{|\mathcal{T}|} \mid t_1 < t_2 < \dots < t_{|\mathcal{T}|}\}$ contains all timestamps throughout $\mathcal{G}$, $V_T = \{v_1, v_2, \dots, v_{N_T}\}$ represents the nodes and $X_T = \{\phi(v, t) \mid v \in V_T, t \in T\}$

denotes node attributes, with $\phi(\cdot)$ mapping nodes to their attributes.

Continuous-time dynamic graphs employ two principal methodologies to capture temporal network dynamics:

1. **Contact sequences:** These record interactions as discrete events at specific timestamps, treating each interaction's duration as negligible. For instance, in an email network, edges represent instantaneous communication between nodes. This approach is also termed a graph stream [19–23]. Formally, the edge set is defined as $E_T = \{((v_i, v_j), t, w_{ij}) \mid v_i, v_j \in V_T, w_{ij} \in \mathbb{R}_+, t \in T\} \subseteq \left(\frac{V_T}{2}\right) \times T \times \mathbb{R}_+$, indicating an edge (v_i, v_j) with weight w_{ij} exists at timestamp t . In Fig. 4 a timestamps t_2 and t_4 mark when the edge (v_3, v_4) was observed, with unequal intervals $[t_1, t_2]$ and $[t_2, t_3]$.
2. **Interval graphs:** These account for the duration of interactions [20], with time intervals explicitly representing how long connections persist. The edge set is defined as $E_T = \{((v_i, v_j), [t_a, t_b], w) \mid v_i, v_j \in V_T, w \in \mathbb{R}_+, t_a, t_b \in T, t_a < t_b\} \subseteq \left(\frac{V_T}{2}\right) \times \left(\frac{T}{2}\right) \times \mathbb{R}_+$. For example, the edge $((v_3, v_4), [t_3, t_4])$ in Fig. 4b that this connection persists from t_3 to t_4 .

Although continuous-time dynamic graphs present greater challenges than discrete-time dynamic graphs when applying traditional static graph methods, they avoid the critical trade-off between information preservation and computational efficiency inherent in discrete-time approaches.

Notation

This survey uses M to represent any temporal variable in temporal networks. The notation M_s^e is a shorthand for variable M in a temporal network from timestamp s to timestamp e . Despite differences between discrete-time and continuous-time dynamic graphs regarding the temporal variable M , this survey unifies them through timestamp indices.

Fig. 3 An example of discrete-time dynamic graphs (DTDG)

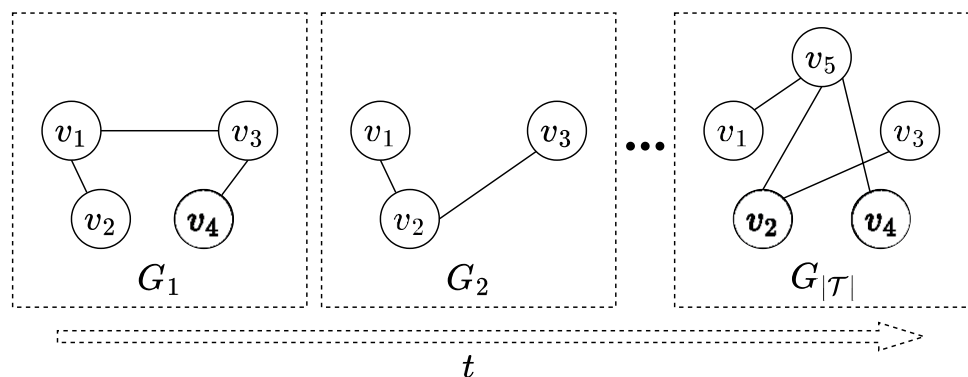
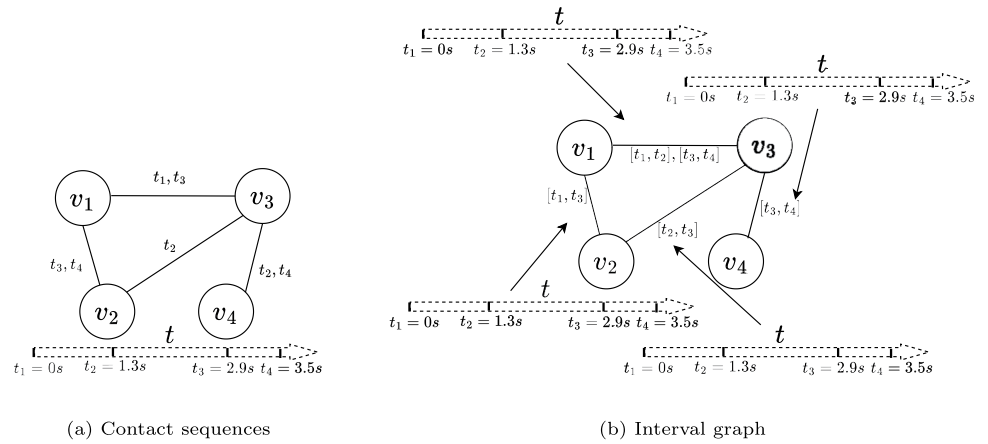


Fig. 4 Continuous-time dynamic graphs (CTDG)



1. Discrete-time Dynamic Graphs: The sequences of snapshots of the variable M from timestamp s to timestamp e are denoted as $M_s^e = (M_s, M_{s+1}, \dots, M_e)$.
2. Continuous-time Dynamic Graphs: The continuous-time representation of the variable M from timestamp s to timestamp e is denoted as M_s^e .

In addition, the rest of this survey uses τ to denote the index of the current timestamp, Δ as the forthcoming timestamps and Δ' as the preceding timestamps.

Inference Tasks

Transductive Tasks

A transductive task (or transductive inference) involves learning and predicting among observed data [24]. In transductive TLP, the model predicts edges between observed nodes. Transductive TLP models f_{TT} use the historical graph $G_{\tau-\Delta'}^{\tau}$ from $(\tau - \Delta')$ to τ and node attributes $X_{\tau-\Delta'}^{\tau+\Delta}$ (if available) from $(\tau - \Delta')$ to $(\tau + \Delta)$ as input to predict the future graph from τ to $(\tau + \Delta)$. The transductive inference process is:

$$\hat{G}_{\tau}^{\tau+\Delta} = f_{\text{TT}}(G_{\tau-\Delta'}^{\tau}, X_{\tau-\Delta'}^{\tau+\Delta}) \quad (1)$$

where $(\Delta \geq 1)$, $(\tau \geq \Delta' \geq 0)$ and $\hat{G}_{\tau}^{\tau+\Delta}$ is the predicted graph. As the node set $\mathcal{V}$ has been observed during training, information is encompassed within $G_{\tau-\Delta'}^{\tau}$.

Inductive Tasks

An inductive task (or inductive inference) pertains to reasoning from observed training data to generalise to unseen data [25]. Inductive TLP methods f_{IT} use the historical graph $G_{\tau-\Delta'}^{\tau}$ from timestamp $(\tau - \Delta')$ to τ , the node attributes $X_{\tau-\Delta'}^{\tau+\Delta}$ (if available) from timestamp $(\tau - \Delta')$ to $(\tau + \Delta)$,

and future nodes $V_{\tau}^{\tau+\Delta}$ from timestamp τ to $(\tau + \Delta)$ as input to predict the future graph $\hat{G}_{\tau}^{\tau+\Delta}$, induced by nodes $V_{\tau}^{\tau+\Delta}$. This is:

$$\hat{G}_{\tau}^{\tau+\Delta} = f_{\text{IT}}(G_{\tau-\Delta'}^{\tau}, X_{\tau-\Delta'}^{\tau+\Delta}, V_{\tau}^{\tau+\Delta}) \quad (2)$$

where $(\Delta \geq 1)$ and $(\tau \geq \Delta' > 0)$. TLP can be categorised into one-step tasks $(\Delta = 1)$ and multi-step tasks $(\Delta > 1)$ based on discrete-time dynamic graphs. Most existing methods primarily address one-step tasks.

Note that all the methods presented in this survey that can deal with induction tasks apply to transductive tasks but the reverse is not necessarily true.

Summary

This section introduced the foundations of temporal networks and the problem formulation of TLP. Temporal networks are modelled as temporal graphs, where the evolution of nodes, edges, and attributes can be described in either discrete-time or continuous-time paradigms, each offering different trade-offs between efficiency and information preservation. TLP extends traditional link prediction by requiring models to capture both spatial and temporal dynamics for forecasting future interactions. We further distinguished between transductive and inductive inference tasks, highlighting their respective assumptions and generalisation scopes. Overall, this background sets the stage for understanding the methodological developments reviewed in the following sections. In particular, the distinction between discrete-time and continuous-time representations, as well as between transductive and inductive inference tasks, provides the foundation for the taxonomy that follows.

Taxonomy

The Framework of the Taxonomy

This survey proposes a taxonomy framework for TLP. It contends that all TLP methods comprise two sub-tasks: representing temporal networks and making predictions, which are processed by two distinct components: the Representation Unit and the Inference Unit. The representation unit captures the network structure and dynamics, whilst the inference unit leverages these representations to forecast future links. As depicted in Fig. 5a, nodes, edges, and attributes are processed by a representation unit to produce inputs that capture dynamic patterns. Subsequently, the inference units process features or embeddings derived from the representation unit to infer future links. In [26], transductive and inductive tasks relate to inference units but not to graph representation.

This survey categorises TLP methods based on their underlying elements, despite many approaches being hybrid, combining different representation and inference units or integrating general machine learning strategies. Figure 5a shows that each method incorporates at least one representation unit and one inference unit. Since temporal networks can be modelled as either discrete-time or continuous-time dynamic graphs, and TLP tasks can be either transductive or inductive, methodologies are classified into four distinct categories based on these combinations, as illustrated in Fig. 5b.

Representation Unit

Temporal networks can be represented as discrete-time or continuous-time dynamic graphs, each requiring different graph representation capabilities to capture spatial and temporal information. Representation unit methodologies represent real-world network data and extract features or latent information.

The representation unit fundamentally encapsulates the spatial dynamics within the temporal networks. As illustrated in Fig. 6, three principal methodologies are identified for the processing of graph dynamics in TLP:

1. **Snapshots:** This approach necessitates directly feeding network snapshots into the predictive model. Because each snapshot can be considered as a static network and structurally depicted via an adjacency matrix $\mathcal{A}$, the Snapshots technique is commonly employed within discrete-time dynamic graphs.
2. **Features Extraction:** This approach entails extracting explicit features from the node, edge, or entire graph levels such as similarity or centrality degree. These features are then used as input for the model. Features extraction is often employed to gather complex information beyond Snapshots to enhance the performance of models.
3. **Latent Variable:** It represents latent variables that echo underlying factors and they are typically used to reduce the dimensionality and suppress noise within temporal networks. These approaches are commonly used when graph dynamics are highly complex and not easily summarised by explicit feats. Discrete-time and continuous-time dynamic graphs employ distinct approaches. However, all methods for continuous-time dynamic graphs fall under latent variables while discrete-time dynamic graphs methods encompass all three types mentioned above.

Table 1 catalogues the representation unit methods referenced in this survey, which are examined thoroughly in “[Review of Representation Units for TLP](#)”. To address TLP, researchers have proposed various representation units that fall into two categories: task-dependent units (specifically designed for TLP) and task-independent units (applicable across diverse downstream tasks in temporal networks). The process of generating latent variables through these

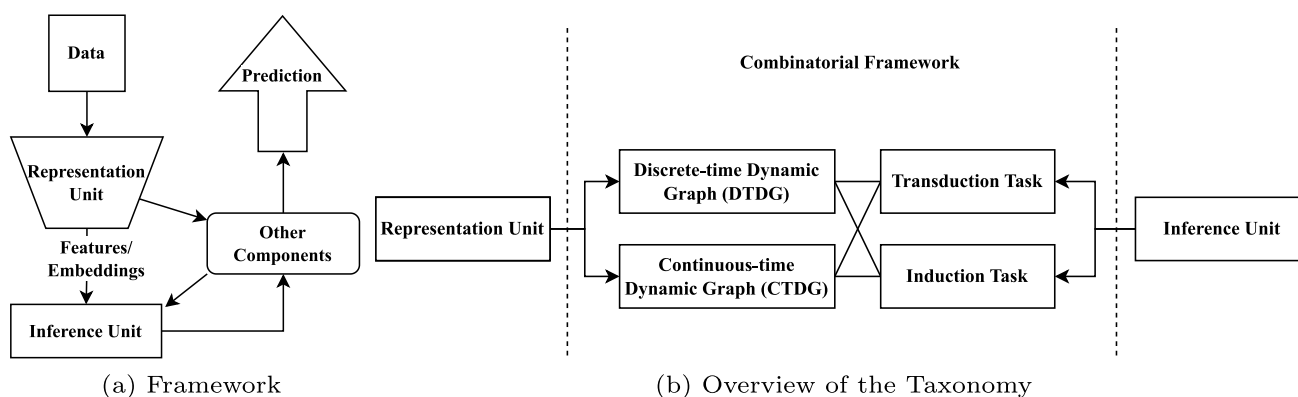


Fig. 5 The framework of the taxonomy

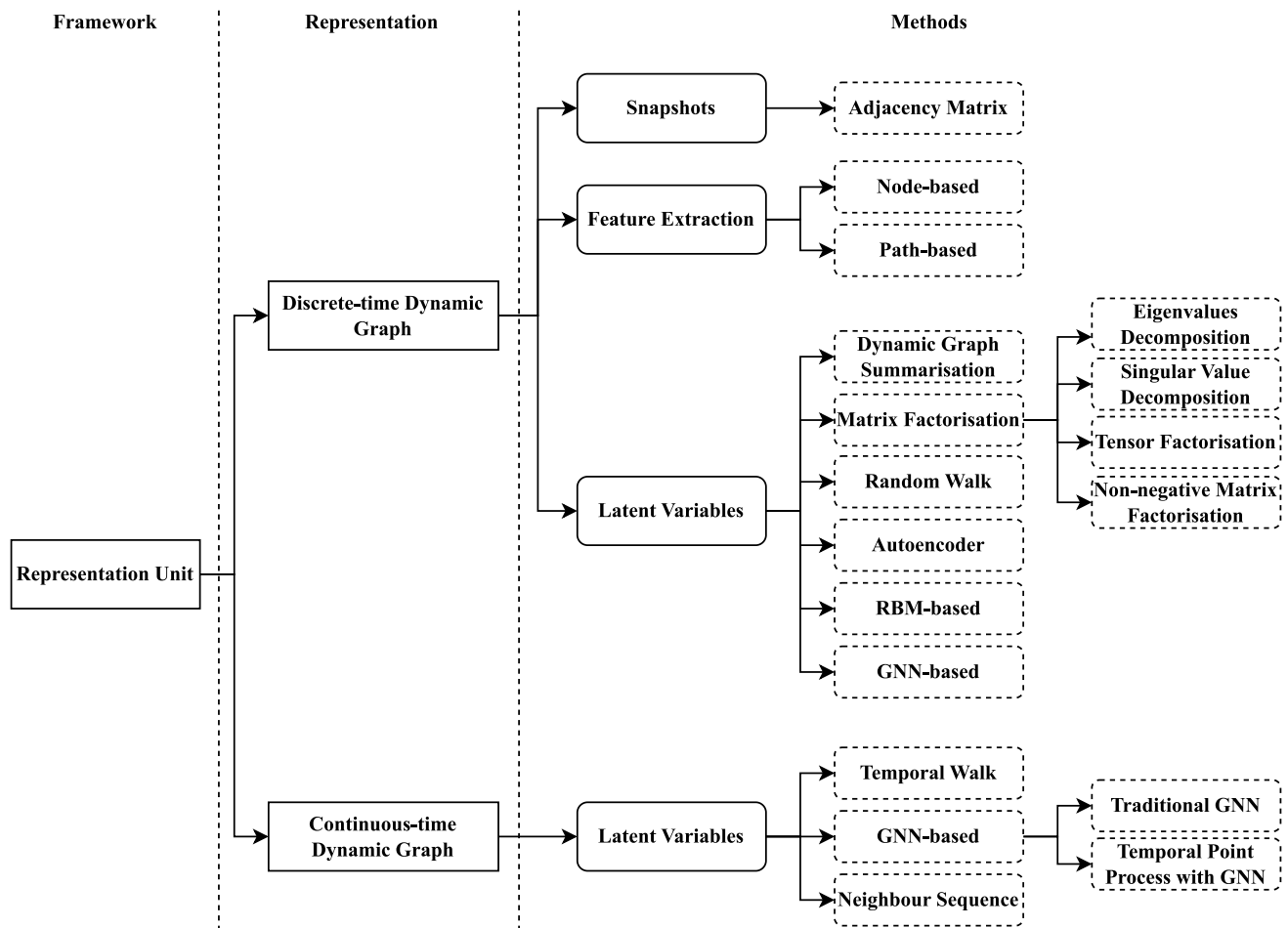


Fig. 6 Taxonomy of representation unit

representation units is broadly termed Dynamic Network Embedding (DNE) [13, 14, 63]. DNE maps nodes to high-dimensional vector representations whilst preserving the dynamic patterns of topology and attributes. Formally, DNE learns a function $f_{\text{DNE}} : \mathcal{V} \mapsto \mathbb{R}^d, d < N$ that projects the node set $\mathcal{V}$ into a lower-dimensional space.

In TLP applications, task-dependent DNE methods are optimised specifically for link prediction, yielding superior performance but limited transferability. In contrast, task-independent DNE methods create versatile embeddings suitable for multiple downstream tasks, including node classification and community detection, though they may deliver comparatively lower performance on TLP tasks.

Inference Unit

Modelling temporal dynamics is related to transductive and inductive tasks as shown in Fig. 7, which demands different abilities to integrate the upstream representation and learn temporal patterns for TLP. Inference units are the methods employed to process graph representations derived from

representation units and learn dynamic patterns of temporal networks for TLP.

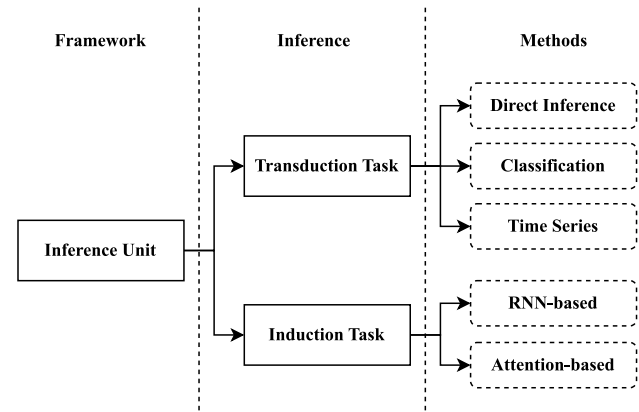
Direct inference is a prevalent approach in representation unit applications, whereby embeddings or features from dynamic network embedding methods directly yield predictions without requiring additional model training. Techniques such as Euclidean Distance or adjacent matrix transformations are commonly employed. The method's straightforward nature makes it particularly popular for one-step tasks.

Moreover, it is important to highlight that several inference methods, including RNN-based and attention-based models, initially designed for inductive tasks, can effectively handle transductive tasks. This shows the adaptability of these models and suggests the potential for their broader application in different TLP scenarios. Table 2 summarises all the methods of inference units which are elaborated in “A Review of Inference Units for TLP”.

Table 1 Summary of methods in representation unit

Methods	Representation	Inference	Representation Unit
Node-based	Discrete	Transductive	Feature Extraction
COMMLP-FULL [27]	Discrete	Transductive	Feature Extraction
CAES [28]	Discrete	Transductive	Feature Extraction
Path-based	Discrete	Transductive	Feature Extraction
DGS [29]	Discrete	Transductive	DGS
ED [30]	Discrete	Transductive	MF
TSVD/CP [31]	Discrete	Transductive	MF
LIST [32]	Discrete	Transductive	MF
TMF [33]	Discrete	Transductive	MF
TBNS [34]	Discrete	Transductive	MF
DeepEye [35]	Discrete	Transductive	MF
GRJMF [36]	Discrete	Transductive	MF
GrNMF [37]	Discrete	Transductive	MF
SNMF-FC [38]	Discrete	Transductive	MF
AM-NMF [39]	Discrete	Transductive	MF
DeepWalk [40]	Discrete	Transductive	Random Walk
Node2Vec [41]	Discrete	Transductive	Random Walk
DynNode2Vec [42]	Discrete	Transductive	Random Walk
SGNE [43]	Discrete	Inductive	Random Walk
SDNE [44]	Discrete	Transductive	Autoencoder
Dyn-VGAE [45]	Discrete	Transductive	Autoencoder
DynGEM [46]	Discrete	Inductive	Autoencoder
ctRBM [47]	Discrete	Transductive	RBM-based
CTDNE [48]	Continuous	Transductive	Temporal Walk
Graph WaveNet [49]	Discrete	Transductive	GNN-based
SDG [50]	Discrete	Transductive	GNN-based
TGN [1]	Continuous	Transductive	GNN-based
TGGDN [51]	Continuous	Transductive	GNN-based
CoDyG [52]	Continuous	Transductive	GNN-based
HTNE [53]	Continuous	Transductive	GNN-based
MTNE [54]	Continuous	Transductive	GNN-based
M ² DNE [55]	Continuous	Transductive	GNN-based
DyGCN [56]	Discrete	Inductive	GNN-based
GCN-MA [57]	Discrete	Inductive	GNN-based
TGAT [58]	Continuous	Inductive	GNN-based
TREND [59]	Continuous	Inductive	GNN-based
GraphMixer [60]	Continuous	Inductive	Neighbour Sequence
DyGFormer [61]	Continuous	Inductive	Neighbour Sequence
FreeDyG [62]	Continuous	Inductive	Neighbour Sequence

GNN Graph Neural Network, *DGS* Dynamic Graph Summarisation, *MF* Matrix Factorisation, *RBM* Restricted Boltzmann Machine

**Fig. 7** Taxonomy of inference unit

Relationship of Representation Unit and Inference Unit

As illustrated in Fig. 5a, data are first transformed by the Representation Unit into features or embeddings that capture structural and temporal dynamics. These representations are then processed by the Inference Unit to produce link predictions, often in conjunction with auxiliary components such as candidate generation or negative sampling [1, 81]. A TLP model, therefore, relies on the alignment between the expressive capacity of its Representation Unit and the predictive capability of its Inference Unit. For example, simple feature extraction in the Representation Unit, such as similarity or centrality measures, is typically paired with lightweight inference techniques such as time series models or distance-based scoring. In contrast, latent states derived from graph neural network encoders or continuous-time event-driven models usually require parametric inference modules, including multilayer perceptrons, recurrent decoders, or attention-based predictors.

It is also important to note that many approaches to Representation Units and Inference Units are rooted in neural network architectures. Graph neural networks, recurrent networks, and attention mechanisms are different forms of neural computation that can be employed in either stage. Consequently, TLP models often integrate these components in diverse ways, combining representation and inference modules into flexible and powerful end-to-end architectures. This neural foundation provides both strong approximation ability and architectural adaptability, allowing researchers to design models that are tailored to specific temporal settings while retaining the capacity to generalise across a wide range of prediction tasks.

Table 2 Summary of methods in inference unit

Methods	Representation	Inference	Representation Unit	Inference Unit
TVRC [29]	Discrete	Transductive	DGS	Classification
GTRBM [64]	Discrete	Transductive	RBM-based	Classification
TSalton [65]	Discrete	Transductive	Node-based	Time Series
SR [66]	Discrete	Transductive	MF	Time Series
TRMF [67]	Discrete	Transductive	MF	Time Series
ARIMA-Kalman [68]	Continuous	Transductive	Snapshots	Time Series
DDNE [69]	Discrete	Transductive	Snapshots	RNN-based
DynGraph-2Vec [70]	Discrete	Transductive	Autoencoder	RNN-based
EvolveGCN [12]	Discrete	Transductive	GNN-based	RNN-based
GCN-GAN [71]	Discrete	Transductive	GNN-based	RNN-based
Network-GAN [72]	Discrete	Transductive	GNN-based	RNN-based
DGNN [23]	Continuous	Inductive	GNN-based	RNN-based
CAW-N [73]	Continuous	Inductive	Temporal Random Walk	RNN-based
GAT [74]	Discrete	Inductive	GNN-based	Attention-based
DySAT [75]	Discrete	Inductive	GNN-based	Attention-based
ASTGCN [76]	Discrete	Inductive	GNN-based	Attention-based
STGSN [77]	Discrete	Inductive	GNN-based	Attention-based
NeiDyHNE [78]	Discrete	Inductive	GNN-based	Attention-based
MAGNA [79]	Continuous	Inductive	GNN-based	Attention-based
DyRep [80]	Continuous	Inductive	GNN-based	Attention-based

GNN Graph Neural Network, *DGS* Dynamic Graph Summarisation, *MF* Matrix Factorisation, *RBM* Restricted Boltzmann Machine, *RNN* Recurrent Neural Network

Temporal Graph Benchmarks

In recent years, significant progress has been made toward standardising evaluation in temporal graph link prediction, most notably through the Temporal Graph Benchmark (TGB) [82] and its extension TGB 2.0 [83] on multi-relational and

heterogeneous temporal graphs. These benchmarks provide datasets, unified preprocessing pipelines, and public leaderboards,¹ enabling systematic comparisons among a wide range of methods.

TGB includes datasets from diverse domains. For example, a dataset on Wikipedia co-editing interactions focuses on predicting which page a user will edit at a given time [84]. Another dataset on Amazon product reviews considers the task of predicting the next product a user will review [85]. A financial dataset based on stablecoin ERC20 transactions requires forecasting the destination address of a transfer, including cases during the Terra Luna collapse [86]. A social media dataset constructed from Reddit reply dynamics predicts whether one user will respond to another [87]. Finally, a dataset on international flights predicts whether a flight will occur between two airports on a specified day [88]. Collectively, these datasets span domains such as collaborative editing, e-commerce, finance, social media, and transportation, highlighting the broad range of real-world applications for TLP.

Both TGB and TGB 2.0 report systematic comparisons of TLP methods, ranging from classical heuristics and static baselines to state-of-the-art temporal graph neural networks such as JODIE, TGN, and CAW. The leaderboard results highlight that while sophisticated temporal models often achieve strong predictive performance, simple memory-based heuristics also remain competitive, especially on large-scale datasets.

Summary

Building on the distinctions introduced in the previous section, this taxonomy frames TLP methods around two complementary units: the Representation Unit and the Inference Unit. The Representation Unit reflects the discrete-time vs continuous-time modelling paradigm, focusing on how temporal networks are represented and encoded. The Inference Unit corresponds to the transductive vs inductive inference setting, determining how predictions are carried out given the learned representations. By combining these two dimensions, we categorise existing methods into four principal groups, providing a coherent structure for understanding these approaches. This design not only ensures consistency with the conceptual foundations of temporal networks but also highlights open challenges, such as bridging hybrid Representation and Inference Units, balancing representational fidelity and efficiency, and enabling models to generalise across diverse TLP scenarios.

¹ <https://tgb.complexdatalab.com/>.

Review of Representation Units for TLP

Discrete-time Dynamic Graphs Methods

Snapshots

The Snapshots approach is a fundamental method employed in the domain of discrete-time dynamic graphs for TLP. This technique directly inputs network temporal adjacency matrices into the inference unit. The snapshots are treated as a sequence of static networks and are structurally represented by adjacency matrices. This method is effective for tasks that resemble time-series inference. The direct use of snapshots simplifies the representation and processing of dynamic graphs, making it an efficient choice for many application scenarios.

Feature Extraction

Node-based Similarity Approaches

Entities tend to form new connections with highly similar to themselves. Node neighbourhoods are crucial in similarity calculation. Building on this observation, researchers have developed neighbour-based methods that utilise neighbourhood topological information from discrete snapshots for TLP. In this survey, $\Gamma(v_i^\tau)$ represents the set of neighbours of the node v_i^τ , and $|\Gamma(v_i^\tau)|$ denotes the degree of the node v_i^τ (the number of its neighbours).

Common Neighbours (CN) [89] It measures the number of shared neighbours of two nodes v_i^τ and v_j^τ . The higher the number of common neighbours, the higher the likelihood of a link between the nodes:

$$\text{CN}(v_i^\tau, v_j^\tau) \equiv |\Gamma(v_i^\tau) \cap \Gamma(v_j^\tau)| \quad (3)$$

Preferential Attachment (PA) [90] This index suggests that the likelihood of a new connection between two nodes is proportional to the product of their degrees. It is often used in growing scale-free networks:

$$\text{PA}(v_i^\tau, v_j^\tau) \equiv |\Gamma(v_i^\tau)| \cdot |\Gamma(v_j^\tau)| \quad (4)$$

Jaccard Coefficient (JC) [91] The JC measures the common neighbours by the total number of unique neighbours of both nodes:

$$\text{JC}(v_i^\tau, v_j^\tau) \equiv \frac{|\Gamma(v_i^\tau) \cap \Gamma(v_j^\tau)|}{|\Gamma(v_i^\tau) \cup \Gamma(v_j^\tau)|} \quad (5)$$

Sorensen Index (SI) [92] The SI compares the number of common neighbours to the sum of the degrees of both nodes. It is more robust than the Jaccard Coefficient against outliers:

$$\text{SI}(v_i^\tau, v_j^\tau) \equiv \frac{2|\Gamma(v_i^\tau) \cap \Gamma(v_j^\tau)|}{|\Gamma(v_i^\tau)| + |\Gamma(v_j^\tau)|} \quad (6)$$

Cosine Similarity (CS) [93] The CS measures the similarity between two nodes by calculating the cosine of the angle between their neighbour vectors:

$$\text{CS}(v_i^\tau, v_j^\tau) \equiv \frac{|\Gamma(v_i^\tau) \cap \Gamma(v_j^\tau)|}{\sqrt{|\Gamma(v_i^\tau)| \cdot |\Gamma(v_j^\tau)|}} \quad (7)$$

Adamic/Adar (AA) [94] AA weights common neighbours with smaller degrees more:

$$\text{AA}(v_i^\tau, v_j^\tau) \equiv \sum_{v_p^\tau \in \Gamma(v_i^\tau) \cap \Gamma(v_j^\tau)} \frac{1}{\log |\Gamma(v_p^\tau)|} \quad (8)$$

Node-based similarity approaches typically serve as the foundation for many community-based TLP models, such as COMMMLP-FULL [27] and CAES [28]. These methods offer high interpretability.

Path-Based Similarity Approaches

Path-based methods provide another perspective for measuring the similarity between nodes in a network. These methods concentrate on the length and quantity of paths between nodes. Unlike node-based approaches, path-based methods view networks as a series of connected paths, enabling the identification of similarities beyond immediate neighbourhoods. Some path-based methods also incorporate random walks to account for the uncertainty and temporal evolution of networks.

In the rest of the survey, $(M)_{a,b}$ and $(\mathcal{M})_{a,b,c}$ denote elements of matrix M or tensor $\mathcal{M}$ in the a -th row, b -th column, and c -th depth, respectively. m_a represents the a -th vector of the matrix M .

Shortest Path (SP) [95] The SP converts the shortest path length between nodes into a similarity measure by taking its negative value. This transformation ensures that shorter paths correspond to higher similarity scores, while longer paths indicate lower similarity.

$$\text{SP}(v_i^\tau, v_j^\tau) \equiv -|d(v_i^\tau, v_j^\tau)| \quad (9)$$

where $d(\cdot)$ is the shortest path between the node pair (v_i^τ, v_j^τ) computed by the Dijkstra algorithm [96].

Local Path (LP) [97] LP makes use of information on local paths between node v_i^τ and node v_j^τ with a 2, 3 and 4-length rather than the nearest paths. LP suggests that 2-length paths should have greater significance than 3-length paths and 3-length paths in relation to 4-length paths. To account for this, an adjustment factor α is applied.

$$\text{LP}(v_i^\tau, v_j^\tau) \equiv A_\tau^2 + \alpha A_\tau^3 + \alpha^2 A_\tau^4, \quad (10)$$

where A_τ^2 , A_τ^3 and A_τ^4 denote adjacency matrices about the nodes having 2, 3 and 4-length distances at timestamp τ , respectively.

Katz Index (KI) [98] The KI measures similarity based on the number of paths of different lengths between two nodes. Shorter paths have larger similarities.

$$\text{KI}(v_i^\tau, v_j^\tau) \equiv \sum_{\ell=1}^{\infty} \beta^\ell |\text{paths}^\ell(v_i^\tau, v_j^\tau)| = \left(\sum_{\ell=1}^{\infty} \beta^\ell A_\tau^\ell \right)_{i,j}^{-1} = (\mathbf{I} - \beta A_\tau)_{i,j}^{-1} - \mathbf{I}_{i,j}$$

where $\text{paths}^\ell(v_i^\tau, v_j^\tau)$ is the set of total ℓ -length paths between nodes v_i^τ and v_j^τ . β is a damping factor, which constrains the path weights and $\mathbf{I}$ is the identity matrix.

Cosine Similarity Time (CST) [99] The CST is a variation of Cosine Similarity:

$$\text{CST}(v_i^\tau, v_j^\tau) \equiv \frac{(L_\tau^\dagger)_{i,j}}{\sqrt{(L_\tau^\dagger)_{i,i} (L_\tau^\dagger)_{j,j}}} \quad (11)$$

SimRank (SR) [100] The SR supposes that two nodes are considered similar if they connect to similar nodes. This method calculates similarity based on the probability that two random walkers, starting from nodes v_i^τ and v_j^τ , will meet at the same node in the future. The walkers move to a random neighbour at each step and the similarity is computed recursively.

$$\text{SR}(v_i^\tau, v_j^\tau) \equiv \begin{cases} 1 & v_i^\tau = v_j^\tau \\ \alpha \cdot \frac{\sum_{v_p^\tau \in T(v_i^\tau)} \sum_{v_q^\tau \in T(v_j^\tau)} \text{SR}(v_p^\tau, v_q^\tau)}{|\Gamma(v_i^\tau)| \cdot |\Gamma(v_j^\tau)|} & v_i^\tau \neq v_j^\tau \end{cases} \quad (12)$$

These feature extraction approaches perform TLP through direct classification of the extracted features. Additionally, these approaches primarily focus on deriving one-step tasks based solely on the current snapshot G_τ . These methods, while effective in certain scenarios, do not fully exploit the dynamics of temporal networks.

Latent Variables

Dynamic Graph Summarisation (DGS)

DGS is primarily a task-independent representation unit used to integrate historical graph snapshots into one comprehensive weighted snapshot. In the context of TLP, DGS plays a critical role in synthesizing historical network dynamics. DGS collapses successive historical snapshots $G_{\tau-\Delta'}^\tau$ into a comprehensive weighted snapshot $\tilde{G}_{\tau-\Delta'}^\tau$ using some kernel functions to aggregate while ensuring that essential properties of the dynamic topology are preserved. The DGS process is given by:

$$\begin{aligned} \tilde{G}_{\tau-\Delta'}^\tau &\equiv \alpha_{\tau-\Delta'} G_{\tau-\Delta'} + \alpha_{\tau-\Delta'+1} G_{\tau-\Delta'+1} + \cdots + \alpha_\tau G_\tau \\ &= \sum_{t=\tau-\Delta'}^{\tau} K(G_t; \tau, \theta) \end{aligned} \quad (13)$$

The work in [29, 101] introduced DGS as a representation unit and compared it to traditional classification models, referred to as inference unit. They built the model based on the successive adjacency matrix A .

The primary distinction among various DGS methods lies in their kernel functions, such as the exponential and linear functions. [102] demonstrated DGS methods' applicability to TLP through direct inference.

Matrix Factorisation (MF)

MF, also known as matrix decomposition, decomposes historical adjacency matrices $A_{\tau-\Delta'}^\tau$ or their transformations to extract latent features of temporal networks. These features can then be combined with other inference units or make direct inferences of future snapshots.

Several matrix factorisation techniques have been applied to TLP with considerable success. These include Eigenvalue Decomposition (ED), Singular Value Decomposition (SVD), Tensor Factorisation (TF), and Non-negative Matrix Factorisation (NMF), all of which are frequently used as task-dependent dynamic network embedding methods for TLP.

Eigenvalue Decomposition (ED) ED [103] utilises spectral graph theory to mine latent variables for tracking and predicting temporal networks. Spectral analysis involves the ED of the Laplacian matrix for each snapshot. In graph theory, each eigenvalue can be seen as a specific frequency revealing particular aspects of the graph's structure, such as its connectivity, robustness, and community structure. For a given timestamp t , the ED can be written as:

$$A_t = U_t \Lambda_t U_t^\top \quad (14)$$

where A_t is the adjacency matrix of a network, U_t is an orthogonal matrix, and $\Lambda_t = \text{diag}(\lambda_1^t, \lambda_2^t, \dots, \lambda_N^t)$ is the eigenvalue diagonal matrix at the timestamp t . Here, N denotes the number of nodes in the network, which also corresponds to the dimension of the adjacency matrix A_t . [30] studied spectral evolution based on ED and took advantage of two successive timestamps to solve TLP.

Singular Value Decomposition (SVD) SVD [103] is a generalised Eigenvalue Decomposition (ED) that extends to non-square matrices. SVD is commonly integrated with spectral analysis or dynamic graph summarisation to generate embeddings for TLP. For a collapsing adjacency matrix, $\bar{A}_{\tau-\Delta'}^\tau$ of dynamic graph summarisation, its SVD form is:

$$\bar{A}_{\tau-\Delta'}^\tau = U \Sigma V^\top \quad (15)$$

where U and V are orthogonal matrices, and Σ is a diagonal matrix containing the singular values. Typically, U is the target embedding of the historical graph, while V and Σ are supplementary matrices.

Truncated SVD (TSVD) TSVD is a variant of SVD that provides a low-rank approximation of the input matrix. Applying a K -rank approximation, the TSVD of $\bar{A}_{\tau-\Delta'}^\tau$ can be given by:

$$\bar{A}_{\tau-\Delta'}^\tau \approx U_K \Sigma_K V_K^\top \quad (16)$$

where U_K and V_K are matrices formed from the first K columns of U and V , respectively, and Σ_K is the principal $K \times K$ submatrix of Σ . The work in [31, 104] leveraged the Katz Index [98] based on the collapsing adjacency matrix to generate a similarity score matrix for TLP.

LIST [32] and TMF [33] both employ SVD to analyse network dynamics. LIST integrates spatial and temporal data for both one-step and multi-step tasks, using gradient descent to optimise its inference unit. This methodology is advantageous for temporal networks with diverse structural properties but may introduce significant computational overhead during the optimisation process. TMF is less computationally demanding, making it more efficient but potentially less effective in networks where spatial relationships are key and only suitable for multi-step tasks.

Tensor Factorisation (TF) TF is derived from SVD. The CanDecomp/Parafact (CP) decomposition is one of the most important TF methods for TLP [105]. The work in [31, 104] examined homogeneous and bipartite graphs, such as user-item networks. These graphs have nodes and edges of the same type, divided into two disjoint sets of size m and n (where $m + n = N$), such that their sum equals the total number of nodes, N . They defined the relationship matrix

R of size $m \times n$, where each element $(R)_{i,j}$ represents the relationship between v_i and v_j :

$$(R)_{i,j} = \begin{cases} 1 & \text{if } v_i \text{ links to } v_j \\ 0 & \text{otherwise} \end{cases} \quad (17)$$

For temporal networks, this relationship matrix becomes R_t at timestamp t , and extends to a tensor $\mathcal{R}^{m \times n \times |\mathcal{T}|}$ across all snapshots.

TBNS [34] utilises tensor factorisation as the representation unit, capturing time series trends in network data. TBNS processes network snapshots into a two-dimensional matrix using exponential smoothing, enabling effective node similarity calculations for link prediction. However, TBNS generates sparse matrices, which may result in storage inefficiencies.

Non-negative Matrix Factorisation (NMF) [36] introduced NMF for TLP. [106] showed that NMF decomposes a non-negative matrix such as an adjacency matrix A_t into two non-negative matrices $U_t \in \mathbb{R}_+^{N_t \times d}$ and $V_t \in \mathbb{R}_+^{d \times d}$. Here d is the dimension of the latent space ($d < N_t$). Their product approximates the original matrix:

$$A_t \approx U_t V_t^\top \quad (18)$$

where U_t is the target embedding matrix. This method is also known as classic bi-factor NMF [106]. It needs to optimise the following Frobenius norm $\|\cdot\|_F$ of each snapshot:

$$\arg \min_{U_t \geq 0, V_t \geq 0} J_{\text{BNMF}} \equiv \|A_t - U_t V_t^\top\|_F^2 \quad (19)$$

GRJMF [36] and AM-NMF [39] comprehensively integrate temporal snapshots with node attributes. GRJMF utilises a tri-factorisation method, which is thorough but computationally intensive. AM-NMF combines matrix factorisation from consecutive snapshots, balancing historical and recent data through a decay factor for TLP.

SNMF-FC [38] and DeepEye [35] prioritise computational efficiency by simplifying the modelling process. SNMF-FC focuses on temporal data through a decayed summation of feature matrices, potentially overlooking spatial details. DeepEye stabilises network dynamics across snapshots with consensus matrices, efficient in networks with gradual changes but possibly missing abrupt shifts.

GrNMF [37] extends SNMF-FC by incorporating graph regularisation to better maintain spatial relationships among nodes. This approach enhances the model's ability to capture both spatial and temporal information albeit with increased computational demands.

In summary, GRJMF and GrNMF are suitable for detailed spatial-temporal analysis but require significant computational resources. SNMF-FC and DeepEye provide efficient alternatives for less dynamic networks, while AM-NMF's adaptable framework makes it versatile for a variety of network analysis scenarios, balancing detail with computational efficiency.

Random Walk

In a static network, a classic random walk is a sequence of nodes where each vertex is randomly chosen from the neighbours of the current node. The random walk starts at an initial vertex and continues by selecting a neighbouring vertex with a certain transition probability and length L to form a walk $W = (v_a^{t(1)}, v_b^{t(2)}, \dots, v_c^{t(L)})$ such that the r -th node $v_i^{t(r)} \in V_t$ and edge $(v_i^{t(r)}, v_j^{t(r+1)}) \in E_t$.

There are static and dynamic approaches for the random walk in discrete-time dynamic graphs for TLP. Static random walk methods model each snapshot independently to generate embeddings, which are then used by the representation units for TLP. Well-known algorithms that apply this approach include DeepWalk [40] and Node2Vec [41, 107]. In contrast, dynamic random walk methods directly model the dynamics of the network such as DynNode2Vec [42] and SGNE [43].

Static Methods DeepWalk [40] and Node2Vec [41] are influential graph analysis methods that utilise random walks to learn node embeddings, drawing inspiration from Skip-gram model [108]. DeepWalk generates node sequences through random walks and employs the Skip-gram model to capture higher-order proximity between nodes, effectively preserving local connectivity patterns. In contrast, Node2Vec extends DeepWalk's methodology by introducing biased random walks that balance breadth-first and depth-first search [109, 110], enabling the model to capture both local and global topological information. This makes Node2Vec particularly adept at exploring complex network structures.

[107] first applied Node2Vec to temporal networks by using Node2Vec on each snapshot independently to generate an embedding sequence for each node. Then, they utilised traditional classifiers as inference units to make TLP, including Logistic Regression [111], Random Forests [112], and Gradient Boosted Regression Trees [113]. However, these static methods have issues with time-consuming computations and consistency.

In summary, Node2Vec offers a more flexible approach than DeepWalk by allowing for adjustable exploration of a node's neighbourhood. However, both methods face scalability issues and computational challenges when applied

to temporal networks, necessitating considerations of efficiency and dynamic consistency in their application.

Dynamic Methods [43] proposed SGNE, extending the Skip-gram algorithm based on DynNode2Vec [42]. While both methods adapt Skip-gram to capture temporal dynamics in discrete-time representation, SGNE introduces two key innovations: a decomposable loss function for learning representations of new nodes, and a selection mechanism that identifies the most influential nodes affected by network evolution. The focus on influential nodes potentially enables SGNE to achieve higher accuracy in TLP, particularly in networks where specific nodes drive structural changes.

Autoencoder

Autoencoder usually consists of encoders and decoders. The encoder transforms inputs into the latent representation space, whereas the decoder maps these embeddings back to obtain reconstructed inputs. Embeddings can be learned by minimising the reconstruction loss function. Nowadays, autoencoders are widely implemented by various kinds of neural networks.

DynGEM [46], SDNE [44], and Dyn-VGAE [45] are three distinct approaches to dynamic node embeddings, each utilising variations of autoencoders to adapt to network evolution. DynGEM leverages a deep autoencoder, incrementally updating embeddings from one snapshot to the next and dynamically adjusting its architecture to accommodate growing network sizes. This flexibility in handling network expansion is advantageous but may lead to increased complexity in the neural network configuration. SDNE [44], a semi-supervised model, reconstructs the adjacency matrix to maintain first-order and second-order proximities, which helps preserve local and global topological features. This dual focus allows SDNE to maintain high fidelity in representing node relationships but at the cost of potentially higher computational demands due to the semi-supervised nature of the loss function. Dyn-VGAE [45] advances this by integrating Variational Graph Autoencoders (VGAE) [114] with Kullback–Leibler divergence [115], optimising embeddings for current snapshots while ensuring temporal consistency. This approach provides a robust framework for temporal coherence but requires careful tuning of hyperparameters to balance immediate embedding accuracy with longitudinal consistency. Overall, each model presents a unique strategy to handle dynamic networks, with varying degrees of complexity and focus on either architectural flexibility, topological fidelity, or temporal stability.

Restricted Boltzmann Machine-based Approaches

The Restricted Boltzmann Machine (RBM) is a deep learning structure originating from the Markov random field. It has two layers: a visible layer $v \in \mathbb{R}^{N \times 1}$ and a hidden layer

$h \in \mathbb{R}^{d \times 1}$. The elements in v and h are stochastic binary variables representing observable data and latent representations [116]. RBM defines a joint distribution over v and h through the energy function:

$$\Pr(v, h) = \frac{\exp\{v^\top W h + a^\top v + b^\top h\}}{\sum_{v, h} \exp\{v^\top W h + a^\top v + b^\top h\}} \quad (20)$$

where $W \in \mathbb{R}^{N \times d}$ is the weight matrix; the bias vectors $a \in \mathbb{R}^{N \times 1}$ and $b \in \mathbb{R}^{d \times 1}$ correspond to v and h respectively. The loss function J_{RBM} aims to minimise the negative log-likelihood:

$$\min_W J_{\text{RBM}}(W; v, h) \equiv -\ln \left(\sum_h \Pr(v, h) \right) \quad (21)$$

ctRBM [47] can capture complex non-linear variations using an exponentially large state space. ctRBM consists of two separate layers of visible units, the historical layer $\tilde{v}$ and the predictive layer v , both fully connected to hidden units h . These layers receive inputs from the historical topology $G_{\tau-\Delta'+1}^\tau$ and the prediction result $\hat{G}_{\tau+1}$ (or the training ground truth), respectively. ctRBM makes a prediction based on the current time window $(\tau - \Delta', \tau]$ and local neighbour. For each node v_i , define $\tilde{v} = ((A_{\tau-\Delta'+1})_{i,:}, \dots, (A_\tau)_{i,:})^\top \in \mathbb{R}^{N\Delta' \times 1}$ and $v = (A_{\tau+1})_{i,:}^\top \in \mathbb{R}^{N \times 1}$. ctRBM uses direct inference to predict the future adjacency matrix $\tilde{A}_{\tau+1}$:

$$(\hat{A}_{\tau+1})_{i,:}^\top = f_{\text{TT}}(G_{\tau-\Delta'}^\tau) \equiv \Pr(v \mid h; \delta) = \sigma(W h + a) \quad (22)$$

Graph Neural Networks-based Discrete-time Approaches

Graph neural networks (GNNs) are a type of neural networks that operate on graph-structured data. As one of the most popular models in recent years, GNNs possess powerful abilities in graph representation learning [117]. Most GNNs are based on the message-passing mechanism. This powerful mechanism consists of memorizing and aggregating the messages of nodes in the network. Therefore, GNNs can capture the dynamics in the temporal networks and update the embeddings of nodes and edges accordingly. Apart from their representation ability, the flexible architectures of GNNs help them integrate with various other dynamic network embedding methods. Thus, GNNs can handle both discrete-time and continuous-time dynamic graphs. Same as random walks, GNNs can also be divided into static and dynamic methods for discrete-time dynamic graphs.

Static Methods Several GNNs have been proposed and achieved impressive results on static networks such as Graph Convolutional Network (GCN) [118], LINE [119] and GraphSAGE [120]. They all generate node embeddings. GCN uses convolutions to aggregate information from node neighbours. LINE preserves first-order and second-order proximities between nodes to learn embeddings. GraphSAGE is based on GCN and uses a sampling technique to handle large graphs.

Spatial-temporal GNNs (STGNNs) are a type of GNN that extends traditional static GNNs to handle temporal networks with spatial information [121, 122]. STGNNs constitute a distinct class of GNNs. These networks model the dynamics by accounting for dependencies between connected nodes. STGNNs jointly capture spatial correlations and temporal dynamics. Thus, STGNNs are widely applied in traffic forecasting [123–125], and epidemic prediction [126].

[49] proposed Graph WaveNet by extending the Convolutional Neural Network (CNN) [127, 128] to an STGNN. The Graph WaveNet consists of a spatial convolution layer and a temporal convolution layer. The spatial convolution layer combines a diffusion convolution layer [129] with a self-adaptive adjacency matrix, while the temporal convolution layer adopts a gated version of a dilated causal convolution network [130].

Dynamic Methods Dynamic GNNs are directly designed to handle discrete-time dynamic graphs. There are also several other dynamic GNNs based on the Graph Convolutional Network (GCN). For example, Stochastic Gradient Descent (SGD) [131] combines GCN with PageRank similarity [50, 132, 133].

DyGCN [56] is a typical task-independent dynamic network embedding method based on GCN to address the challenges of temporal networks. The key of DyGCN is to generalise the embedding propagation scheme of GCN to a dynamic setting in an efficient manner to update the node embedding matrix Z_τ . DyGCN assumes the feature matrix X is fixed and the node embeddings are updated according to the change of aggregated message.

GCN-MA [57] uses a GCN as the representation unit. GCN-MA includes the novel Node Representation Node Aggregation Effect algorithm which is a combination of GCN, RNN, multi-head attention mechanisms, enhancing node representation through node degree, clustering coefficient, and neighbour relationships. Thus, GCN-MA has an improved ability to capture global and local temporal patterns.

Critical Comparison

We present head-to-head comparisons along three axes: (1) representation performance, (2) training efficiency and scalability, and (3) robustness.

Dynamic GNNs vs Random Walks

(1) *Performance*: Both Dynamic GNNs and Random Walks can be applied in transductive and inductive learning, but they capture network signals in different ways. Dynamic GNNs combine neighbourhood aggregation with temporal encoding or memory modules, enabling the integration of structural and attribute information and maintaining continuity across time. Random Walks generate node sequences that emphasise higher-order proximity and structural context, which can be advantageous in structure-only graphs but less effective when attributes play a major role in prediction. (2) *Efficiency*: Dynamic GNNs reduce cost through mini-batch sampling and incremental updates on affected subgraphs, maintaining a favourable cost-accuracy balance over long horizons. Random Walks are inexpensive per snapshot (walks $\times$ length) but accumulate overhead from repeated sampling, negative sampling, and alignment across multiple timestamps. (3) *Robustness*: Dynamic GNNs can alleviate local fluctuations through aggregation but may oversmooth or lose fine-grained details in highly dynamic settings. Random Walks are sensitive to window configuration and negative sampling, and without explicit alignment they may drift across snapshots. Nevertheless, the paths they generate often provide more interpretable higher-order structural patterns. In practice, Dynamic GNNs are suitable when attributes and inductive generalisation are central, while Random Walks remain competitive baselines in structure-driven scenarios.

DGS vs Matrix Factorisation

(1) *Performance*: DGS preserves coarse topology and recency but loses event order and higher-order motifs. Matrix Factorisation captures low-rank, multiway, and periodic structure and tends to perform better on periodic or bipartite streams. (2) *Efficiency*: DGS is near-linear and stream-friendly. Matrix Factorisation requires repeated truncated or streamed decompositions as graphs evolve. (3) *Robustness*: DGS smooths noise. NMF with graph regularisation provides parts-based interpretability under sparse and nonnegative regimes. Use DGS as the fast baseline; switch to Matrix Factorisation when periodic or multiway signals matter and decomposition budget exists.

Matrix Factorisation vs Autoencoder and RBM

(1) *Performance*: Autoencoder and RBM models capture nonlinear dynamics and missingness beyond low-rank

structure. Matrix Factorisation excels when dynamics are approximately factorisable or periodic. (2) *Efficiency*: Autoencoder and RBM training is heavier and incremental updates are non-trivial. Matrix Factorisation is also costly but benefits from truncated solvers. (3) *Robustness*: NMF is interpretable and robust under non-negativity constraints. Autoencoder and RBM require careful temporal regularisation and are sensitive to hyperparameters. Prefer Autoencoder or RBM only if nonlinearity or missingness dominates and tuning budget is available.

Dynamic GNNs vs Matrix Factorisation

(1) *Performance*: Dynamic GNNs dominate in inductive and attribute-rich settings and reduce cross-time drift. Matrix Factorisation suits stationary and low-rank temporal patterns. (2) *Efficiency*: Dynamic GNNs support local updates on affected subgraphs. Matrix Factorisation incurs repeated global solves. (3) *Robustness*: Dynamic GNNs leverage features for denoising. NMF remains the interpretable option when non-negativity is required.

Random Walks vs DGS

(1) *Performance*: Random Walks capture higher-order structural context. DGS sacrifices such detail for stability. (2) *Efficiency*: Both are lightweight per snapshot or window. Random Walks accumulate cost over many timestamps, whereas DGS streams with rolling kernels. (3) *Robustness*: Random Walks are sensitive to sampling and alignment and exhibit cross-time drift. DGS is stable but less expressive. Choose DGS for stable streaming baselines.

Continuous-Time Dynamic Graphs Methods

Latent Variables

Graph Neural Networks-based Continuous-time Approaches

Traditional GNN TGAT [58] extends the dynamic network embedding ability of GAT [74] via Function Time Encoding. GAT targets static settings and does not consider the temporal dynamics between neighbours. To process continuous-time dynamic graphs, the time features used in TGAT are derived from concepts based on Bochner's Theorem [134] to map time to $\mathbb{R}^d$. Then, $\Phi(t)$ is concatenated with node embedding for GAT to solve TLP. However, TGAT cannot maintain the historical state of the nodes.

TGN [1] memorises the history information to generate node embeddings for continuous-time dynamic graphs. The model comprises several modules, including Memory, Message Function, Message Aggregator, Memory Updater, and Embedding. The Memory module stores a vector for each

node representing the node's history in a compressed format at a certain timestamp. Other modules implement the message-passing mechanism to generate temporal node embedding. Therefore, TGN performs better than TGAT.

TGGDN [51] employs a group affinity matrix to model both local and long-distance interactions within networks, incorporating a transformer architecture for temporal data processing and enhanced interpretability. CoDyG [52] introduces a co-attention mechanism alongside a temporal difference encoding strategy to effectively capture evolving correlations between node pairs over time.

These approaches represent distinctive methodologies for modelling temporal graph dynamics, with TGGDN focusing on group-level interactions and broader network patterns, whilst CoDyG emphasises the fine-grained temporal evolution of pairwise node relationships.

Temporal Point Process with GNN The temporal point process is a probabilistic generative model for continuous-time event sequences, which involves a stochastic process whose realisation consists of a list of discrete events localised in time [135]. Assuming an event happens within a tiny period $[t, t + dt)$, the temporal point process represents the conditional probability of this event given historical events as $\lambda(t)dt$. The Hawkes process [136] is a widely-used temporal point process method for TLP. The Hawkes process is described by the equation [137]:

$$\lambda(t) = \mu(t) + \int_{-\infty}^t \kappa(t-s)dE(s) \quad (23)$$

where the conditional intensity function $\lambda(t)$ represents the instantaneous rate of event occurrence at timestamp t ; $\mu(t)$ represents the base intensity, which indicates the rate at which spontaneous events occur at timestamp t ; $\kappa(t-s)$ is the kernel, modelling the time decay of past events' influence on the current intensity; $E(t)$ signifies the number of events up to timestamp t . This process incorporates self-exciting and past influence mechanisms of events, allowing it to effectively capture the complex dependencies and dynamics in continuous-time event sequences [138, 139].

[53] introduced HTNE, which captures network evolution by modelling how past interactions influence future connections through an attention mechanism that weighs historical interactions based on temporal proximity.

Building upon this foundation, M²DNE [55] incorporates both micro-dynamics (individual interactions) and macro-dynamics (subgraph changes) through a dual-attention

mechanism that enhances TLP by balancing direct interactions with overarching network evolution patterns.

MTNE [54] addresses limitations in previous approaches by modelling network dynamics through triad motif evolution and the Hawkes process. This method captures mesoscopic structural patterns neglected by HTNE and M²DNE, which primarily focus on direct neighbour interactions and broad network changes, respectively.

TREND [59] combines the Hawkes process with Temporal Graph Networks to model both event dynamics and node dynamics simultaneously. It characterises each edge formation as an event with properties determined by the participating nodes at specific timestamps.

These approaches contribute different methodological perspectives to continuous-time dynamic graph embedding: HTNE focuses on temporal relevance through attention mechanisms, M²DNE addresses micro and macro network dynamics, MTNE captures mesoscopic structural patterns via motif evolution, and TREND combines event and node dynamics. Each method offers distinct techniques for modelling temporal network relationships, providing various analytical frameworks for understanding complex network behaviours.

Temporal Walk

A temporal walk extends the concept of random walks [40, 41] to continuous-time dynamic graphs. Formally, a temporal walk is defined as a sequence of nodes $W_T = (v_a^{(1)}, v_b^{(2)}, \dots, v_c^{(L)})$ with length L during time domain T , where the r -th node $v^{(r)} \in V_T$, the r -th edge $((v^{(r)}, v^{(r+1)}), t^{(r)}) \in E_T$, and each timestamp $t^{(r)} \in T$. Crucially, timestamps must follow a temporal ordering (either ascending or descending) to properly capture network dynamics.

Unlike standard random walks, temporal walks feature irregular time intervals between steps whilst respecting the temporal ordering of connections, making them particularly suitable for continuous-time dynamic graphs.

CTDNE [48] leverages temporal walks for dynamic network embedding. The method first generates these walks using edge selection based on uniform, exponential, or linear distributions, then applies the Skip-Gram model [140] to learn the resulting embeddings. As a task-independent dynamic network embedding approach, CTDNE produces representations applicable across various temporal network tasks.

Neighbour Sequence

A neighbour sequence is an ordered history of a node's one-hop interactions over time, where each element records the neighbour identity and the interaction timestamp. Focusing on one-hop contexts preserves salient local evidence for link formation while avoiding neighbourhood expansion, and naturally supports inductive inference because unseen nodes can be encoded as new sequences without reprocessing the entire graph.

GraphMixer [60] Recent studies specify how the sequence is tokenised, how time is encoded, and how the sequence is read out for prediction. GraphMixer [60] follows a minimalist pipeline: each interaction (i, j, t) (often represented relative to the prediction time as $(i \rightarrow j, \Delta t)$) is mapped by an MLP link encoder into a token-level embedding; the node representation is then obtained by permutation-invariant mean pooling over the recent one-hop tokens, optionally with simple absolute or relative time encodings. Dispensing with message passing and multi-hop expansion yields near-linear training and inference complexity in sequence length, and training typically uses binary cross-entropy on time-respecting positives and negatives. Empirically, this lightweight mixing retains most predictive signal in structure-dominated settings and scales well on large graphs.

DyGFormer [61] targets longer-range temporal dependencies and cross-neighbour interactions within the same one-hop history. It augments tokens with a neighbour co-occurrence encoding so that the model captures which neighbours tend to appear together within short temporal windows, reducing sensitivity to local ordering noise. The sequence is then processed by a Transformer [141] with relative time encodings and a causal or time-respecting mask, allowing attention to aggregate evidence across distant events in a single pass. This improves multi-horizon inference when predictive signals span many past interactions or when correlations among multiple neighbours are important. Although attention has quadratic cost in context length, windowed or sparse attention and batched training over nodes help maintain scalability.

FreeDyG [62] builds on the same one-hop formulation but explicitly models temporal regularities through frequency encoding. Interaction timestamps within a node's sequence are projected onto sinusoidal or learnable frequency bases (for example, Fourier-style features over Δt), producing band-specific responses that reveal periodic or seasonal rhythms. These frequency features are concatenated with token embeddings and consumed by a lightweight encoder, enabling the model to align interactions across different rates and to amplify periodic components that are predictive

of future links. Reported results show that frequency-aware tokens stabilise training under irregular sampling and bring consistent gains on streams with daily or weekly cycles, while keeping the one-hop footprint and the inductive interface unchanged.

Overall, treating one-hop temporal interactions as sequences provides a scalable and inductive substrate. Mixing-based encoding as in GraphMixer is suited to local and recent signals on large graphs, attention-based encoding as in DyGFormer better models long-range and multi-source temporal context, and frequency-aware encoding as in FreeDyG leverages periodic structure when present. In all cases, time-respecting construction and evaluation of neighbour sequences are essential to avoid leakage and to ensure faithful assessment of temporal generalisation.

Critical Comparison

We compare continuous-time representation units along three axes: (1) representation performance, (2) training efficiency and scalability, and (3) robustness.

Dynamic GNNs vs Neural Temporal Point Processes

(1) *Performance*: Dynamic GNNs integrate structural aggregation with temporal encoding or memory to generate evolving embeddings, supporting inductive generalisation and alleviating representation drift. Temporal point processes directly model event intensities and are more effective when precise timing signals and excitation effects drive network evolution. (2) *Efficiency*: Dynamic GNNs update incrementally through mini-batch sampling on affected sub-graphs, maintaining efficiency in long streams. Temporal point processes require likelihood estimation over events, with complexity depending on kernel design; simple decay kernels are efficient, while learned kernels increase cost. (3) *Robustness*: Dynamic GNNs adapt well in attribute-rich settings but may oversmooth in highly dynamic environments. Temporal point processes provide explicit interpretability through base rates and decay terms, but are sensitive to kernel mis-specification. Hybrid approaches combine the robustness of GNN embeddings with the timing fidelity of temporal processes.

Temporal Walks vs Dynamic GNNs

(1) *Performance*: Temporal walks respect chronological order and provide strong structure-driven baselines, but remain transductive since new nodes require re-walks. Dynamic GNNs offer inductive embeddings using neighbourhood and attribute information. (2) *Efficiency*: Temporal walks are inexpensive per snapshot but accumulate

cost across many timestamps. Dynamic GNNs amortise computation through sampling and incremental updates. (3) *Robustness*: Temporal walks are sensitive to sampling design but yield interpretable path structures. Dynamic GNNs are more stable with attributes and explicit temporal regularisation.

Neighbour-sequence Models vs Dynamic GNNs

(1) *Performance*: Neighbour-sequence models compress local temporal context and perform strongly with simple encoders, extended by attention or frequency encoding for long-term or periodic signals. They trade higher-order structural coverage for simplicity, while general GNNs recover multi-hop dependencies through message passing. (2) *Efficiency*: Sequence models avoid neighbourhood explosion and scale well; attention-based variants are heavier than basic encoders. General GNNs rely on sampling, which becomes costly in high-degree graphs. (3) *Robustness*: Frequency encodings improve robustness in seasonal streams and offer frequency-domain interpretability. General GNNs remain preferable when multi-hop dependencies or rich attributes dominate.

Temporal point processes or hybrids are preferable when timing and excitation patterns are central. Dynamic GNNs are suitable for inductive deployments with rich attributes. Group-based models target community diffusion, pairwise models address fine-grained dynamics, temporal walks serve as efficient structure-only baselines, and neighbour-sequence models provide scalable local context, escalating to full GNNs when multi-hop or attribute signals are essential.

Summary

This section reviewed representation units for TLP, following the taxonomy introduced earlier. In the discrete-time setting, Representation Units range from simple snapshots and feature-based similarity measures to more advanced latent-variable models such as matrix factorisation, random walks, autoencoders, RBMs, and GNNs. These methods progressively trade interpretability for expressiveness, capturing increasingly complex temporal dynamics. In the continuous-time setting, Representation Units rely heavily on latent-variable approaches, including GNN-based models with memory or time encoding, temporal point processes, temporal walks, and neighbour sequence methods. Collectively, these approaches highlight two complementary insights: discrete-time methods emphasise structural clarity via snapshot-based modelling, while continuous-time methods prioritise fine-grained temporal fidelity. Open challenges

remain in improving scalability to large networks and balancing task-specific optimisation with transferability across downstream applications.

A Review of Inference Units for TLP

Transductive Inference Methods

Direct Inference

Direct inference is a widely used technique to solve TLP. This approach is characterised by its simplicity and directness, allowing the embeddings or features generated through dynamic network embedding methods to be utilised immediately without the need for further training in other inference units. Commonly implemented techniques in this context include matrix factorisation, dynamic graph summarisation, GTRBM [64], and ctRBM [47]. These techniques are valued for their straightforward application in one-step tasks, making them one of the most popular choices in the field.

Classification

TLP can be framed as a binary classification problem, where the aim is to predict the presence or absence of a link between two nodes in the future. Most methods that fall into this category rely on supervised learning. These classifiers usually utilise features or embeddings obtained from representation units for TLP. Various classifiers have been employed for link prediction including Logistic Regression [111], Support Vector Machine (SVM) [142], and Decision Trees (DT) [143].

TVRC [29] combines dynamic graph summarisation as the representation unit with Weighted Relational Bayes Classifiers [144] as the inference unit. TVRC feeds dynamic graph summarisation inputs into Weighted Relational Bayes Classifiers (RBC) [144] or Weighted Relational Probability Trees (RPT) [145] for TLP. RBC extends naive Bayes classifiers and RPT extends standard probability estimation trees to a relational setting. Both RBC and RPT incorporate weights, derived from the collapsed snapshots of dynamic graph summarisation, to better model temporal networks.

GTRBM [64] uses RBM as the representation unit and Gradient Boosting Decision Tree (GBDT) [146] as the inference unit to form the model. GTRBM used the same RBM as ctRBM but the embeddings are input into the GBDT which has better inference ability than Bayes Classifier. As

a result, GTRBM provides better performance than ctRBM which relies on direct inference for TLP.

Time Series

Temporal networks can be conceptualised as time-series data, allowing TLP problems to be approached through established time-series methodologies. These approaches vary based on their applicability to discrete-time or continuous-time dynamic graphs.

Methods for Discrete-time Dynamic Graphs

Classical time series techniques including Auto-Regressive (AR), Auto-Regressive Moving Average (ARMA), Auto-Regressive Integrated Moving Average (ARIMA) [147, 148], and Vector Auto-Regressive (VAR) [149], form the backbone of many TLP models when combined with various representation units.

Feature extraction methods incorporate similarity measures such as Common Neighbours, Adamic-Adar and Jaccard Coefficient [89, 91, 94] with time series analysis. For example, TSaltan [65] integrates ARIMA as its inference unit to predict node centrality, thereby enhancing TLP adaptability dynamically.

Matrix factorisation approaches such as SimRank (SR) [66] and TRMF [67] apply time series principles to dynamic matrix representations. SR combines spectral analysis with ARMA to predict future eigenvectors for graph reconstruction, employing K -rank approximation as Eq. (16) to address computational challenges in large networks. Similarly, TRMF pairs Non-negative Matrix Factorisation with AR to predict temporal embeddings, effectively adapting matrix factorisation to dynamic contexts.

These methods offer different strengths: feature extraction approaches like TSaltan directly incorporate temporal dynamics into node similarity metrics, whilst matrix factorisation methods capture the structural information. Computational requirements vary significantly, with spectral methods requiring dimensional simplifications and NMF methods offering more straightforward implementation but potentially less structural detail.

Methods for Continuous-time Dynamic Graphs

Continuous-time methods expand the traditional Discrete-time approaches such as AR and ARMA to Continuous-time AR (CAR) [150] and Continuous-time ARMA (CARMA) [151]. Moreover, [68] combined ARIMA and Kalman filtering [152] as the ARIMA-Kalman model for continuous-time dynamic graphs. The ARIMA initialises the Kalman measurement and state equations. The model combines the linear pattern-capturing strengths of ARIMA and the

adaptive noise reduction ability of Kalman filtering. Therefore, ARIMA-Kalman improves flexibility and forecasting accuracy of TLP.

Critical Comparison

We compare major inference-unit families for TLP along three axes: (1) inference performance, (2) training efficiency and scalability, and (3) robustness. Representation units supply features or embeddings; inference units map these to future link probabilities or event intensities.

Classification-based Inference vs Time-series Inference

(1) *Inference performance*: Classification learns a discriminative mapping from latent features and attributes to link probabilities, capturing cross-feature interactions; it is effective when rich attributes complement topology and when the prediction target is horizon-agnostic. Time-series inference encodes temporal dependence (lags, seasonality, regime persistence) and is advantageous when dyad-level histories are informative and horizon-specific forecasts are required. (2) *Efficiency and scalability*: Shallow classifiers train in near-linear time in samples and feature dimension; stronger ensembles improve fit at higher but still batch-friendly cost. Time-series models are light per series but become costly when fitted per dyad at scale; hybrids that couple factorisation with AR-type inference add decomposition overhead. (3) *Robustness*: Probabilistic classifiers and trees offer transparent decision logic and support calibration; ensembles are stronger but less interpretable. Time-series models provide parameter-level interpretability and principled uncertainty, but are sensitive to misspecification and concept drift.

Classification-based Inference vs Direct Inference

(1) *Inference performance*: Direct inference tied to the representation objective (e.g., reconstruction scores, similarity thresholds) is simple but may underfit decision boundaries. Using the same embeddings with supervised classifiers typically yields higher predictive accuracy by learning task-specific boundaries. (2) *Efficiency and scalability*: Direct inference has minimal training overhead and is attractive for rapid deployment. Classification adds training cost but can reuse cached embeddings and supports incremental retraining. (3) *Robustness*: Direct inference inherits the inductive bias of the representation and is easy to audit, yet can be brittle under class imbalance. Classification enables probability calibration, threshold tuning, and feature attribution, at the expense of additional hyperparameters.

Time-series Inference vs Direct Inference

(1) *Inference performance*: Time-series approaches outperform direct inference when temporal patterns drive link formation and when multi-horizon or irregular-timestamp forecasts are needed. Direct inference does not explicitly capture serial dependence and may miss lagged effects. (2) *Efficiency and scalability*: Direct inference is the cheapest option at run time. Time-series fitting introduces per-dyad estimation costs; shared-parameter formulations (e.g., vector or state-space models) mitigate but do not eliminate scaling challenges. (3) *Robustness*: Time-series models are statistically interpretable and, with filtering, tolerate missing or noisy observations; however, they degrade under unmodelled structural breaks. Direct inference is simple and stable when dynamics are mild but provides limited temporal diagnostics.

Inductive Inference Methods

Recurrent Neural Networks-Based Approaches

Recurrent Neural Networks (RNNs) refer to a class of neural networks where connections between nodes can form cycles, enabling output from some nodes to influence subsequent input to the same nodes. This allows RNNs to model sequential data and process the dynamics of temporal networks.

However, traditional RNNs suffer from vanishing and exploding gradients. Consequently, RNNs may be unable to capture long-range data dynamics dependencies. To overcome these issues, Long Short-Term Memory (LSTM) [153] and Gated Recurrent Unit (GRU) [154] networks were developed based on RNNs to handle long-term learning issues. RNNs could also be divided into discrete-time and continuous-time methods.

Methods for Discrete-time Dynamic Graphs

DDNE [69] uses adjacency matrices to create time-specific embeddings. DDNE employs two Gated Recurrent Units (GRUs) to analyse these embeddings, capturing the network's evolution over time. This method enables the generation of advanced embeddings and dynamic inferences using Multi-Layer Perceptions (MLP). The dual GRU — one processing data from past to present, the other in reverse — allows DDNE to understand temporal dependencies within networks comprehensively. This facilitates accurate TLPs, showcasing DDNE's capability to harness network dynamics effectively.

DynGraph2Vec [70] uses an autoencoder as the representation unit and RNN as the inference unit. The method processes adjacency matrices from previous time steps, employing a Multi-Layer Perceptron (MLP) as the encoder to transform these matrices into embeddings. This is followed by an LSTM-based decoder focused on TLP.

DynGraph2Vec introduces variants such as DynGraph2VecAE, DynGraph2VecAERNN, and DynGraph2VecRNN to cater to different requirements and optimisation strategies.

The encoding process leverages MLP to convert the series of adjacency matrices over time into a sequence of node embeddings, effectively capturing the graph's evolving structure. The decoder, on the other hand, uses an LSTM layer fed by the generated embeddings to predict future adjacency matrices as TLP. This approach allows DynGraph2Vec to understand and anticipate the dynamic changes in graph structures, facilitating accurate embeddings for temporal networks.

EvolveGCN [12] utilises a Graph Convolutional Network (GCN) module as the representation unit to learn node embeddings in temporal networks. EvolveGCN employs a GRU to update the weights of the GCN for inference at each timestamp, allowing the model to capture the latent dynamic patterns of the temporal networks. For TLP, it employs a Multi-Layer Perceptron (MLP) that uses the final GCN-generated embeddings to predict future connections, which efficiently captures and predicts dynamics in temporal networks.

GCN-GAN [71] leverages GCN as the representation unit and LSTM as the inference unit under the architecture of a Generative Adversarial Network (GAN) [155–157]. The GAN contains a generator and a discriminator. The generator contains the representation unit and the inference unit to make TLP, while the discriminator is an auxiliary structure to refine the prediction results from the generator. Similarly, NetworkGAN [72] improved the GCN-GAN model with matrix factorisation-based representation units.

Methods for Continuous-time Dynamic Graphs

DyGNN [23] tracks the effects of new edges in temporal networks, blending GNN and LSTM to generate and update embeddings. This model captures both direct interactions and the influence on adjacent nodes through Continuous-time LSTM. A subsequent LSTM layer, equipped with attention and decay mechanisms, spreads the updates across the network, ensuring dynamic and precise embeddings. DyGNN excels in reflecting network changes and improving TLP.

CAW-N is an anonymised variant of the temporal walk, capturing motif evolution in temporal networks. It operates on the premise that nodes with interacting motif structures over time have a higher likelihood of forming links. CAW-N employs temporal walks as representation units to track events between node motifs chronologically, then utilises RNNs for prediction.

Attention-Based Approaches

Attention-based models refer to a set of methods that can mimic cognitive attention, allowing the models to focus on

the important parts of the data while diminishing less relevant parts [158]. These models, such as the global or local attention [159], self/multi-head attention [141], Simple Neural Attentive Learner (SNAIL) [160] and Cardinality Preserved Attention (CPA) [79], have gained popularity in many fields including Computer Vision and Natural Language Processing and have achieved state-of-the-art results in various tasks. One well-known model in graph learning is the Graph Attention Network (GAT) [74]. GAT uses GNNs as the representation unit to process the graph structure and attention mechanism to improve the updating process of node embeddings.

Methods for Discrete-time Dynamic Graphs

DySAT [75] employs a multi-layer GAT [141] as its representation unit alongside a self-attention module for inference. The inductive nature of attention models enables DySAT to generalise to previously unseen nodes, contributing to the increasing popularity of attention-based approaches in temporal graph analysis.

Building on this foundation, ASTGCN [76] and STGSN [77] further develop spatial-temporal modelling techniques. ASTGCN implements distinct attention modules to capture various temporal patterns through spatial-temporal mechanisms and convolution techniques. Similarly, STGSN utilises temporal attention mechanisms to model network evolution from both spatial and temporal perspectives, enhancing the interpretability of dynamic interactions.

NeiDyHNE [78] extends these concepts to heterogeneous networks by integrating neighbourhood interactions with node attributes. It combines a hierarchical structure attention module for analysing node features with a convolutional temporal attention module that captures evolutionary patterns. This integrated approach enables effective management of complex dynamics in heterogeneous networks, improving predictive performance for future connections.

Methods for Continuous-time Dynamic Graphs

Whilst many approaches employ continuous-time GNNs with attention mechanisms for transductive tasks, alternative methods can directly embed continuous-time dynamic graphs using attention models, such as those based on the deep learning Hawkes process described earlier by Eq. (23).

MAGNA [79] extends the attention mechanism from GAT [141] by computing preliminary node embeddings using 1-hop attention matrices, then incorporates multi-hop neighbours through summed powers of these matrices. The model aggregates node features weighted by attention values and processes them through a feed-forward neural network to generate embeddings for TLP.

DyRep [80] takes a different approach based on temporal point processes, modelling temporal networks through two evolving processes: the association process (capturing global network growth) and the communication

process (representing local information propagation). DyRep employs the Hawkes process as its representation unit and applies attention mechanisms to compute edge embeddings for TLP.

Critical Comparison

We compare RNN-based and attention-based inductive inference units along three axes: (1) inference performance, (2) training efficiency and scalability, and (3) robustness. Representation units supply features or embeddings; the inference unit maps these to future link probabilities for unseen nodes or edges.

RNN-based Inference vs Attention-based Inference

(1) *Inference performance*: RNN-based inference captures order-sensitive and short-to-medium temporal dependencies via gating, which is effective when dyad-level histories and recent interactions dominate the predictive signal. Attention-based inference aggregates variable-range contexts in a single step and can integrate multi-hop and heterogeneous cues, which is advantageous when long-range dependencies or cross-source interactions drive link formation. (2) *Efficiency and scalability*: RNNs process sequences sequentially and are less parallelisable; computational cost grows linearly with sequence length and with hidden width. Attention is more parallelisable on modern hardware but its cost grows with the square of the attended context; sparse or locality-aware attention reduces this overhead, whereas windowed RNNs reduce memory but retain sequential latency. (3) *Robustness*: RNNs have compact parameterisation and stable behaviour on moderate sequence lengths, but may under-represent long-range effects even with gated units. Attention exposes contribution weights that support attribution across time and neighbours, yet may overfit under sparse supervision without appropriate regularisation and carefully designed temporal encodings.

Summary

This section reviewed inference units for TLP, aligned with the taxonomy introduced earlier. Transductive inference methods, including direct inference, classification, and time-series models, rely on observed node sets and are effective for one-step predictions but struggle to generalise beyond the training domain. Inductive inference methods, by contrast, employ deep learning architectures such as RNN-based and attention-based models, enabling them to generalise to unseen nodes and capture long-range temporal dependencies. Together, these approaches reflect the core distinction between transductive and inductive tasks established in the background section. Key open challenges include improving the scalability of deep inductive models,

reducing the generalisation gap between transductive and inductive settings, and developing unified inference frameworks that integrate seamlessly with diverse representation units.

Variations of the TLP Problem

This section explores extensions of TLP, from undirected homogeneous temporal networks to more complex variants, including directed, heterogeneous and hypergraph temporal networks, along with their applications. All network definitions are founded on discrete-time dynamic graph principles.

Temporal Directed Networks

A directed temporal network features links that evolve over time with specific directionality between nodes [161, 162]. In a temporal network $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{T}, \mathcal{X})$ where $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$, the edge (v_i^t, v_j^t) indicates a connection from node v_i^t to node v_j^t , distinct from edges in weighted homogeneous attribute temporal networks. These networks model phenomena including rumour or disease propagation via social networks [163, 164], ad hoc message passing [165] and public transport dynamics [166].

Temporal Knowledge Graphs

A Temporal Knowledge Graph is a special type of knowledge graph that incorporates time-varying information into the graph, representing the dynamics of entities and relationships in the knowledge graph [167]. In Temporal Knowledge Graphs, methods for TLP need to consider both the knowledge and evolution of networks [8, 168]. By incorporating time-aware representation learning models, Temporal Knowledge Graphs can reason missing temporal facts and relationships [167, 169]. Various models have been proposed for TLP in Temporal Knowledge Graphs [170, 171] or combined with some Natural Language Processing techniques such as contrastive learning [172] or temporal logic [172].

Temporal Heterogeneous Networks

A heterogeneous temporal network is a network whose nodes and edges are of different types and change over time [173]. Given a heterogeneous temporal network $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{T})$, $\mathcal{V} = \bigcup_{i=1}^n \mathcal{V}_i$ represents n types of nodes, and $\mathcal{E} = \bigcup_{j=1}^m \mathcal{E}_j$ denotes m types of edges. Examples of heterogeneous temporal networks include complex social networks [174] and recommendation systems [175].

Collective Link Prediction refers to a set of methods used for predicting the probability of new relationships forming between nodes in heterogeneous temporal networks [176, 177], and it is often based on attention models [178–180].

Temporal Hypergraph Networks

A temporal hypergraph network is a network whose edges can join any number of nodes and changes over time [181]. Temporal hypergraph networks can represent and analyse complex real-world systems such as economic transactions or transportation [182]. Recently, they have drawn attention and were studied on the TLP problem achieving good performance based on GNNs and attention-based models [183, 184].

Large-scale Temporal Networks

Scalability remains a fundamental challenge for TLP, as real-world networks are often both large-scale and high-frequency. Event-driven models with memory and message passing, such as TGN [1], maintain node states and typically combine temporal neighbour truncation or sampling to bound computation, which makes them practical for long streams. Sequence-projection approaches such as JODIE [84] introduce time-consistent batching strategies that preserve temporal order while improving training efficiency. System-level frameworks like TGL [185] further push scalability by providing specialised temporal data layouts and parallel sampling mechanisms, enabling training on billion-scale temporal graphs. For real-time and high-rate scenarios, APAN [186] decouples inference from graph querying to avoid expensive online multi-hop expansion, substantially reducing latency. These developments illustrate how scalable TLP methods rely on bounding historical dependencies, adopting efficient temporal data structures, and rethinking system-level design to cope with the complexity of modern large-scale temporal graphs.

Multi-Modal Temporal Networks

A multi-modal temporal network is a temporal graph in which nodes or edges are associated with additional modalities such as text, images, or audio that evolve over time. Unlike traditional attribute temporal networks, multi-modal temporal graphs capture richer contextual dynamics by linking structural evolution with heterogeneous content. Research in this direction has been motivated by multi-modal knowledge graphs, where combining structural signals with textual or visual information improves link prediction performance [187]. Recent benchmarks such as UKnow [188] and TGB 2.0 [83] provide large-scale datasets and protocols with temporally grounded multimodal annotations, enabling systematic evaluation. In addition, dynamic text-attributed graph benchmarks (DTGB) demonstrate that evolving textual semantics can serve as strong temporal cues for predicting future links, while also revealing challenges in effectively fusing linguistic and structural dynamics [189]. From a modelling perspective, two main paradigms have emerged. The first is event-aware fusion, which employs time-aligned encoders and cross-modal attention to jointly capture structural evolution and modality-specific signals with temporal sensitivity [189]. The second

is pretraining-based approaches, where graph-text corpora such as MMKG or UKnow are leveraged with cross-modal contrastive learning to initialise temporal graph encoders, later fine-tuned for TLP [190]. Despite these advances, TLP in multi-modal networks still faces open challenges, including modality asynchrony, robustness under missing modalities, and efficient streaming updates. Addressing these issues will be critical for advancing TLP models that can operate under realistic multi-modal and temporally evolving conditions.

Real-world Applications

Temporal characteristics, as well as domain constraints, critically influence how TLP is designed and evaluated across different real-world domains. In transportation networks, traffic flows are governed by spatial topology and physical constraints, and modelling must account for spatio-temporal dependencies. Systems often leverage spatial-temporal graph models that encode connectivity and temporal evolution to forecast future interactions or traffic. For example, deep learning methods such as T-GCN, a fusion of Graph Convolutional Networks with GRUs, effectively capture both spatial (road network) and temporal dependencies in urban traffic forecasting [191]. Broader reviews show that spatio-temporal graphs are widely used in transportation for real-time flow prediction, control, and optimization tasks [192]. In biological systems, such as dynamic protein-protein interaction networks, the interactions are sparse, noisy, and temporally irregular. Continuous to discrete methods mitigate this by modelling temporal smoothness and detecting evolving complexes in dynamic protein-protein interaction networks [193]. Broad-temporal-network surveys underscore the importance of dynamic modelling and temporal awareness in computational biology [194]. Overall, these domains illustrate how TLP must align temporal modelling with domain-specific constraints, such as incorporating spatial adjacency and flow dynamics in transportation and handling irregularity and uncertainty in biological data.

Future Research Directions

TLP represents a vital area of research within complex network science and graph representation learning. This section explores emerging challenges and unresolved questions in the field.

Building New Models for TLP Based on Different Representation Units and Inference Units

In light of the novel framework introduced by this survey, which is characterised by its composite nature, there is a compelling argument for the development of new models for TLP that harnesses the potential of innovative combinations. This framework, by its ability to amalgamate diverse

techniques and methodologies, opens up the possibilities for tackling future TLP challenges in novel and efficacious ways. The essence of this approach lies in its flexibility and adaptability, encouraging the exploration of uncharted territories within the domain of TLP. By integrating different representation units and inference units, this composite framework sets the stage for a holistic and nuanced understanding of network dynamics. As such, not only does it broaden the horizon for TLP research but also serves as a beacon for future investigations, guiding them towards developing solutions that are both innovative and tailored to the complicated temporal network evolution.

Enhancing Explainability in TLP Models

A pivotal future direction lies in improving explainability for TLP models, which are often viewed as black boxes. Despite their predictive success, these models often provide little insight into why a particular link is predicted, which limits trust, transparency, and practical deployment. Existing approaches in graph neural network explanation provide some preliminary foundations. For example, GNNExplainer [195] generates explanations by identifying a compact subgraph and a subset of node features most relevant to a prediction, while [196] analyse explanation mechanisms and show that, under certain conditions, explanations can be approximated via edge gradients, with equivalence to occlusion-based methods in linear settings. However, these methods were designed for static GNNs and do not directly account for temporal dynamics. This gap underscores the challenge: explanations in TLP must capture not only structural rationales but also the timing and evolution of interactions. Future research could adapt to the temporal domain by developing event-aware or sequence-sensitive explanation schemes.

Studying TLP on Continuous-time Dynamic Graphs in Complex Networks

TLP has been a classical task on weighted homogeneous attribute temporal networks for a long time. However, real-world temporal networks are often much more complex. They can be represented as directed, multi-layer, heterogeneous, or hypergraph networks. These complex networks pose challenges in TLP as their structure and evolution patterns are more intricate than the weighted homogeneous attribute temporal networks. Hence, this field remains challenging and requires novel techniques for such complex networks.

Practical Data Challenges in Temporal Graphs

Another crucial direction concerns the data quality issues inherent in temporal graphs. Real-world datasets are often noisy, contain missing or incomplete records, and exhibit irregular or asynchronous time intervals, all of which may distort the learned dynamics and degrade prediction reliability. Addressing these challenges requires developing

representation and inference strategies that are robust to imperfect inputs. One promising line is graph structure learning (GSL), which jointly optimises graph topology and prediction objectives to mitigate noise and missing edges. For example, GTGIB [81] demonstrates how integrating GSL with information bottleneck principles can improve robustness by filtering redundant signals while preserving essential temporal dependencies. In parallel, graph augmentation has recently been explored in temporal graphs. Adaptive augmentation strategies dynamically adjust the strength of temporal and structural perturbations to enhance robustness in temporal graph networks [197]. Unlike GSL, which directly optimises the topology, augmentation focuses on regularising learning by exposing models to varied but coherent graph views; its application in temporal graphs is still at an early stage. Future research should also investigate denoising methods that can filter spurious interactions and temporal models capable of handling irregular event spacing. Progress in these directions would substantially enhance the robustness and applicability of TLP under realistic conditions, where imperfect and heterogeneous temporal data are the norm rather than the exception.

Further Exploration of the Graph Dynamics Mechanism

Recent studies have examined mechanisms including Micro- and Macro-dynamics [55] and Motif structures [54, 198–200]. Nevertheless, a deeper understanding of complex network dynamics and how these mechanisms influence temporal network properties, such as varying speeds, remains elusive. Additionally, developing innovative approaches that leverage these mechanisms to more effectively capture evolution patterns in temporal networks is crucial.

TLP Models in Large-scale Complex Networks

In recent years, the volume of data has been growing exponentially and the development of large models has been progressing rapidly [201, 202]. For instance, social networks are becoming increasingly complex, encompassing various data types. Consequently, constructing a general, large-scale graph prediction model is increasingly promising. Such a model would have a strong ability in multi-task scenarios with domain knowledge, including recommendation systems, knowledge reasoning, and epidemic spreading. Future research in this area should focus on designing scalable and robust TLP models capable of processing large-scale and complex datasets and combining few-shot learning, transfer learning, and other advanced techniques to improve performance.

Pretrained and Foundation-style Temporal Graph Models

A recent trajectory in temporal graph learning shifts from task-specific training toward large-scale pretraining and foundation-style transfer. Encoders are first optimised with self-supervised objectives, such as masked reconstruction of time-conditioned structures or generative forecasting, on long unlabeled streams, and then adapted to TLP, improving data efficiency and cross-domain transfer [203]. In parallel, long-context Transformer backbones have been scaled to encode neighbour sequences and extended histories, strengthening multi-horizon inference when long-range dependencies matter [204]. Beyond single-graph pretraining, cross-network pretraining aims at zero-shot or few-shot adaptation on unseen graphs; recent results report positive transfer and outline scaling behaviours that point toward foundation-style models [205].

Conclusion

This survey introduces a novel taxonomy for Temporal Link Prediction (TLP) that distinguishes between representation units and inference units, providing a structured framework for analysing existing approaches. Through this lens, we have systematically reviewed and classified the literature, revealing the diverse methodological combinations employed across the field. The survey has also examined advanced TLP applications in directed, heterogeneous and hypergraph temporal networks, whilst identifying promising research directions including model explainability, complex network dynamics, and scalable solutions for large-scale networks.

The taxonomy framework presented here offers significant value to researchers by clarifying the fundamental components of TLP methods and highlighting unexplored combinations that could yield performance improvements. By emphasising the separation between representation and inference mechanisms, this work facilitates more targeted methodological innovations. Furthermore, our examination of the evolution from traditional feature-based approaches to advanced neural architectures provides a comprehensive roadmap for the field's development. We anticipate this structured perspective will stimulate new research and accelerate progress towards more effective, interpretable, and versatile TLP models capable of addressing the complex challenges inherent in temporal networks.

A. List of Acronyms

List of acronyms in the survey (Table 3).

Table 3 List of acronyms in the survey

Acronym	Full form
AA	Adamic/Adar
AM-NMF	Adaptive Multiple Non-negative Matrix Factorization
ARIMA	Autoregressive Integrated Moving Average
ARIMA-Kalman	Autoregressive Integrated Moving Average - Kalman filter
ASTGCN	Attention-based Spatial-Temporal Graph Convolutional Recurrent Network
CAES	Community-Aware Evolution Similarity
CAW-N	Causal Anonymous Walks Network
COMMLP-FULL	Community-enhanced Link Prediction in Dynamic Networks
CN	Common Neighbours
CoDyG	Correlation-enhanced Dynamic Graph learning
CP	CANDECOMP/PARAFAC
GRJMF	Graph Regularized Joint Matrix Factorization
CS	Cosine Similarity
CST	Cosine Similarity Time
CTDG	Continuous-time Dynamic Graph
CTDNE	Continuous-Time Dynamic Network Embedding
ctRBM	Continuous-time Restricted Boltzmann Machine
DeepEye	Dynamic Networks Based on Non-negative Matrix Factorization
DeepWalk	Online Learning of Social Representations
DDNE	Deep Dynamic Network Embedding
DGS	Dynamic Graph Summarisation
DGNN	Dynamic Graph Neural Network
DNE	Dynamic Network Embedding
DTDG	Discrete-time Dynamic Graph
Dyn-VGAE	Dynamic Joint Variational Graph Autoencoders
DyGCN	Dynamic Graph Convolutional Network
DyGFormer	Dynamic Graph Transformer
DynGraph2Vec	Dynamic Graph Embedding Model which Captures Temporal Dynamics
DynGraph2VecAE	Dynamic Graph Embedding Model with Autoencoder Architecture
DynGraph2VecAERNN	Dynamic Graph Embedding Model with Autoencoder and Recurrent Neural Network
DynGraph2VecRNN	Dynamic Graph Embedding Model with Recurrent Neural Network
DyRep	Representation Learning Framework for Dynamic Graphs
DySAT	Dynamic Self-Attention Network
DynGEM	Deep Embedding Method for Dynamic Graphs
DynNode2Vec	Dynamic Node2Vec
ED	Eigenvalue Decomposition
EvolveGCN	Evolving Graph Convolutional Networks for Dynamic Graphs
FreeDyG	Frequency Enhanced Continuous-Time Dynamic Graph
GCN	Graph Convolutional Network
GCN-GAN	Graph Convolutional Network - Generative Adversarial Network
GCN-MA	Graph Convolutional Network Long Short-term Memory Network and Multi-head Attention
GNN	Graph Neural Network
Graph WaveNet	Deep Graph Neural Network for Generating Waveforms
GrNMF	Graph Regularised Non-negative Matrix Factorisation
GTRBM	Gradient Boosting Decision Tree Temporal Restricted Boltzmann Machine
HTNE	Hawkes Process-based Temporal Network Embedding
JC	Jaccard Coefficient
KI	Katz Index
LIST	Link Prediction Model with Spatial and Temporal Consistency
LP	Local Path
MAGNA	Multi-hop Attention Graph Neural Network
MF	Matrix Factorisation
M _{DNE}	Temporal Network Embedding Method with Micro- and Macro-Dynamics
MTNE	Motif-preserving Temporal Network Embedding
NeiDyHNE	Neighbor-enhanced Dynamic Heterogeneous Attributed Network Embedding Method

Table 3 (continued)

Acronym	Full form
NetworkGAN	Deep Generative Framework to Tackle the Challenging Temporal Link Prediction
NMF	Non-negative Matrix Factorisation
Node2Vec	An Algorithmic Framework for Learning Continuous Feature Representations for Nodes in Networks
PA	Preferential Attachment
RBM	Restricted Boltzmann Machine
RU	Representation Unit
SDG	Simplified and Dynamic Graph Neural Network
SGNE	Skip-gram based Network Embedding
SI	Sorensen Index
SNMF-FC	Symmetric Non-negative Matrix Factorisation - Feature Collapsing
SP	Shortest Path
SVD	Singular Value Decomposition
SR	SimRank
STGSN	Spatial-Temporal Graph Social Network
TBNS	Tensor-Based Node Similarity
TF	Tensor Factorisation
TGGDN	Temporal Group-aware Graph Diffusion Network
TGN	Temporal Graph Networks
TGAT	Temporal Graph Attention Network
TLP	Temporal Link Prediction
TMF	Temporal Matrix Factorisation
TREND	TempoRal Event and Node Dynamics
TRMF	Temporal Regularised Matrix Factorisation
TSalton	Temporal Salton Index
TSVD	Truncated Singular Value Decomposition
TVRC	Time Varying Relational Classifier

B. Glossary of terms

See Table 4.

Table 4 Glossary of terms and acronyms used in this survey

Term/acronym	Definition
Temporal Network	A network whose structure and properties evolve over time, representing interactions between components of a dynamic system
Temporal Graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{T}, \mathcal{X})$	A formal representation of a temporal network, where $\mathcal{V}$ is the set of nodes, $\mathcal{E}$ the edges, $\mathcal{T}$ the time domain, and $\mathcal{X}$ the node attributes
Snapshot	A static graph representation of a temporal network at a specific timestamp
DTDG (Discrete-time Dynamic Graph)	A temporal graph represented as a sequence of snapshots over discrete timestamps, simplifying continuous dynamics into discrete intervals
CTDG (Continuous-time Dynamic Graph)	A temporal graph representation that preserves continuous timestamps of events, avoiding discretisation
Contact Sequence	A CTDG representation where interactions are recorded as instantaneous events at specific timestamps (also called a graph stream)
Interval Graph	A CTDG representation where edges have durations, i.e., each interaction persists over an interval $[t_a, t_b]$
TLP (Temporal Link Prediction)	The task of predicting future links in a temporal network given historical interactions up to a current timestamp τ
Transductive Task	Inference restricted to observed nodes, predicting future edges between them based on historical data
Inductive Task	Inference that generalises to unseen nodes, predicting links including new nodes not observed during training
Representation Unit	Component of a TLP model that captures temporal network structure and dynamics
Inference Unit	Component of a TLP model that leverages RU outputs to forecast future links

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